EXACT WEIGHT CONES OF GOREN-OORT STRATA IN HILBERT MODULAR VARIETIES

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Allowed standard prerequisites (editorial).

- Original arithmetic geometry and number theory prerequisites. Original complete bibliography and source-local definitions retained.

1. Introduction

Let F be a totally real field of degree $d > 1$ over $\mathbb{Q}$, $\mathcal{O}_F$ its ring of integers, and p a prime which is unramified in F. Recall that the weight of a mod p (geometric) Hilbert modular form is a d-tuple of integers $\kappa = \sum_{\beta\in\mathbb{B}} k_\beta e_\beta \in \mathbb{Z}^{\mathbb{B}}$, where $\mathbb{B} = \mathrm{Hom}(\mathcal{O}_F, \overline{\mathbb{F}}_p)$. Let σ denote the Frobenius automorphism acting on $\mathbb{B}$ by composition on the right. In [2], we introduced the notion of the minimal cone of weights for mod p Hilbert modular forms:

$$\mathcal{C}^{\min} = \{\kappa = \sum_{\beta\in\mathbb{B}} k_\beta e_\beta \in \mathbb{Q}^{\mathbb{B}} : pk_\beta \geq k_{\sigma^{-1}\beta}, \forall \beta \in \mathbb{B}\},$$

and proved that for every mod p Hilbert modular eigenform f, there is an eigenform with the same q-expansion which has weight in $\mathcal{C}^{\min}$. In fact, we proved a stronger result: every mod p Hilbert modular form that is not divisible by any partial Hasse invariants has weight in $\mathcal{C}^{\min}$. An immediate consequence of this result is that the cone of weights of mod-p Hilbert modular forms, i.e., the $\mathbb{Q}^{\geq 0}$-cone generated by the weights of nonzero Hilbert modular forms, is generated by the weights of the partial Hasse invariants, namely, it is the Hasse cone:

$$\mathcal{C}^{\mathrm{Hasse}} = \{\kappa \in \mathbb{Q}^{\mathbb{B}} : \kappa = \sum_{\beta\in\mathbb{B}} x_\beta h_\beta \text{ such that } x_\beta \in \mathbb{Q}^{\geq 0}, \forall \beta \in \mathbb{B}\},$$

where, $h_\beta = -e_\beta + pe_{\sigma^{-1}\beta}$ is the weight of the partial Hasse invariant $\mathbf{h}_\beta$. In [3], we extended these results to the general case where p may be ramified in F.

As part of a program to study the cone of weights of automorphic forms on Shimura varieties, Goldring and Koskivirta [6] also showed, using different methods, that when p is unramified in F, the cone of weights of Hilbert modular forms equals the Hasse cone. One question that arises in the work of Goldring and Koskivirta is to determine Shimura varieties for which the cone of weights is generated by the weights of the relevant Hasse invariants. While, as mentioned above, this is the case over a Hilbert modular variety, it fails to hold over some of its Goren-Oort strata. The central idea in this work is that the generators of the cone of weights should be sought among pullbacks of Hasse invariants of certain Shimura varieties to which the space of interest maps. We will make this precise.

Let $X/\overline{\mathbb{F}}_p$ denote a Hilbert modular variety of level prime to p, and for $\kappa = \sum_{\beta\in\mathbb{B}} k_\beta e_\beta$, let ω^κ be the corresponding automorphic line bundle. Recall that if $\pi : \mathcal{A} \to X$ is the universal abelian scheme over X, then $\pi_*\Omega_{\mathcal{A}/X} \cong \oplus_{\beta\in\mathbb{B}}\omega_\beta$, where, ω_β is a line bundle, $\mathcal{O}_F$ acts on it via β, and

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$\omega^\kappa = \otimes_{\beta\in\mathbb{B}}\omega_\beta^{k_\beta}$. Recall further the partial Hasse invariant $\mathbf{h}_\beta \in H^0(X,\omega^{h_\beta})$ defined in [1]. For any $T\subseteq\mathbb{B}$, the Goren-Oort stratum X_T is defined by the vanishing of partial Hasse invariants $\mathbf{h}_\beta$, for $\beta\in T$. By the cone of weights of X_T, denoted $\mathcal{C}(X_T,\omega^\bullet)$, we mean the $\mathbb{Q}^{\geq 0}$-cone generated by those κ for which $H^0(X_T,\omega^\kappa)\neq 0$. The collection of the Hasse invariants of X_T is given by $\{\mathbf{h}_\beta : \beta\notin T; \mathbf{b}_\beta : \beta\in T\}$, where, for $\beta\in T$, $\mathbf{b}_\beta$ is a nowhere vanishing section of weight $b_\beta = e_\beta + pe_{\sigma^{-1}\beta}$ on X_T, corresponding to the isomorphism $\omega_\beta\cong\omega_{\sigma^{-1}\beta}^{-p}$. In particular, we always have an inclusion of $\mathbb{Q}^{\geq 0}$-cones

$$\langle h_\beta : \beta\notin T; \pm b_\beta : \beta\in T\rangle \subseteq \mathcal{C}(X_T,\omega^\bullet). \tag{1.1}$$

Goldring and Koskivirta [6] showed that the above inclusion is an equality for certain "admissible" strata, a result that also follows from our work in [2], and asked the question if this is the case for all strata. In the first part of this paper, we show that the inclusion is proper for all "non-admissible" strata, and determine the cone of weights for all strata. In fact, we show that every generator of $\mathcal{C}(X_T,\omega^\bullet)$ is obtained by pulling back a Hasse invariant over another stratum $X_{T'}$, via a map $X_T\to X_{T'}$. Let us make this more explicit. For simplicity of presentation, we will assume p is inert in F, and consider only $T\subsetneq\mathbb{B}$. In [8], Tian and Xiao prove that for such T, there is a $T\subseteq\tilde{T}$ of even cardinality, such that X_T is isomorphic to a $(\mathbb{P}^1)^{|\tilde{T}-T|}$-bundle over a quaternionic Shimura variety associated to the quaternion algebra $B_{\tilde{T}}$ that is ramified exactly at $\tilde{T}$, where $\tilde{T}$ is considered as a subset of $\mathrm{Hom}(F,\mathbb{R})$. In particular, this implies that if $T\subseteq T'\subseteq\mathbb{B}$ are such that $\tilde{T}=\tilde{T}'$, then there is a map[1] $\pi_{T,T'}: X_T\to X_{T'}$ which makes X_T a $(\mathbb{P}^1)^{|T'-T|}$-bundle over $X_{T'}$. We prove that $\mathcal{C}(X_T,\omega^\bullet)$ is generated by the weights of the pullbacks of the Hasse invariants of $X_{T'}$ under $\pi_{T,T'}$ as T' varies subject to the above condition. More precisely, for the restricted choice of T as above, we have

$$\mathcal{C}(X_T,\omega^\bullet) = \langle f_\beta(T) : \beta\in\mathbb{B}-T; \pm b_\beta : \beta\in T\rangle,$$

where $f_\beta(T) = -e_{\sigma^{n_\beta}\beta} - (-p)^{n_\beta}e_\beta$, where n_β is the smallest $i>0$ such that $\sigma^i\beta\notin\tilde{T}$, if $\tilde{T}\neq\mathbb{B}$, and $n_\beta = d$ if $\tilde{T}=\mathbb{B}$. When $\tilde{T}=T$, i.e., when X_T is isomorphic to a quaternionic Shimura variety associated to $B_{\tilde{T}}$, this cone equals the cone of weights generated by the weights of the Hasse invariants of X_T (as well as those of the quaternionic Shimura variety). This is the admissible case referred to above. In all other cases, the inclusion 1.1 is proper.

In the second part of the article, we study the question of minimal weights of forms over strata. One motivation for this work has been to clarify what a minimal cone of weights should mean for more general Shimura varieties, and how such minimal cones can be calculated. In [2], the definition of the minimal cone was motivated by considerations arising from the relationship between Hilbert modular forms and Galois representations as in [5]. In this work, we show how the minimal cone of a stratum X_T can be defined using a partial subset of the generators κ of $\mathcal{C}(X_T,\omega^\bullet)$, for which $H^0(C,\omega^\kappa)$ is 1-dimensional for any connected component C of X_T. Starting with this definition, we go on to explicitly calculate the minimal cones of all strata, and prove that these minimal cones play the same role vis-à-vis the minimal weights of Hecke eigenforms over the strata. To account for the (prime-to-p Hecke action), in the final part of this article, we study the same questions over Shimura varieties associated to $\mathrm{Res}_{F/\mathbb{Q}}\mathrm{GL}_2$ (as opposed to the related PEL Shimura varieties implicitly used in the discussion above). We will make this more precise below.

[1]These maps could also be constructed directly using Dieudonné module theory.

Let $X'/\overline{\mathbb{F}}_p$ denote the Shimura variety associated to $\mathrm{Res}_{F/\mathbb{Q}}\mathrm{GL}_2$ of a level prime to p. Again, for any $T \subset \mathbb{B}$, there is a stratum X'_T given by the vanishing of the partial Hasse invariants indexed by $\beta \in T$. In this setting, the weight of a form defined over any stratum is a tuple (λ, κ), where $\kappa = \sum_\beta k_\beta e_\beta$, $\lambda = \sum_\beta \ell_\beta e'_\beta$ are elements of $\mathbb{Z}^{\mathbb{B}}$. Defining the cone of weights $\mathcal{C}(X'_T, \delta^\bullet\omega^\bullet)$ to be the $\mathbb{Q}^{\geq 0}$-cone generated by weights of nonzero forms on X'_T, we show that

$$\mathcal{C}(X'_T, \delta^\bullet\omega^\bullet) = \langle f_\beta : \beta \in \mathbb{B} - T; \pm b_\beta : \beta \in T; \pm e'_\beta : \beta \in \mathbb{B}\rangle$$

We further define and explicitly describe the minimal cone of weights for X'_T. Finally, we prove that for any form $\mathbf{f}$ of weight (κ, λ) on X'_T which is an eigenform for the prime-to-p Hecke algebra, there is a from $\mathbf{f}_0$ on X'_T which has the same Hecke eigenvalues as $\mathbf{f}$, such that the weight of $\mathbf{f}_0$ lies in the minimal cone of X'_T.

While, in [2, 3], we used the explicit description of the minimal cone of X to identify the cone of weights on X, in this work, we first determine the cone of weights for all strata in X, and then use that to find an explicit description of the minimal cones. We propose this as a strategy for describing minimal cones over more general Shimura varieties, in combination with the methods developed by Goldring, Koskivirta and their collaborators.

As stated earlier, every stratum X_T is Hecke-equivariantly isomorphic to a flag space over some quaternionic Shimura variety. We would like to mention that, while we don't pursue this in the article, the exact methods of this work can be used to determine cones of weights and minimal cones of flag spaces over quaternionic Shimura varieties more generally, (and, in particular, those of the full flag spaces). For example, given $T \subset \mathbb{B}$, the cone of weights of the full flag space over a Shimura variety associated to $B_{\tilde{T}}$ is generated by all $f_\beta(T)$ for $\beta \in \mathbb{B}$, and the explicit description of the minimal cone is similar to those obtained in this paper.

Finally, we note that Gabriel Micolet [7] has recently generalized the main result of [8] to the case where p may be ramified in F, using an approach as in [4]. Using the constructions of [7], we expect the results of this paper can be generalized to the case where p is ramified in F.

1.1. **Acknowledgements.** We are grateful to Wushi Goldring, Jean-Stefan Koskivirta, and Deding Yang for several helpful conversations around the themes of this paper. This research also benefited from discussions sparked by a workshop on Serre weight conjectures held at King's College London in March 2024, sponsored by the Heilbronn Institute for Mathematical Research.

2. Hilbert modular varieties

Let F be a totally real field of degree $d > 1$, $\mathcal{O}_F$ its ring of integers, and $\mathfrak{d}_F$ its different. Let p be a prime number which is unramified in F. For a prime ideal $\mathfrak{p}$ of $\mathcal{O}_F$ dividing p, let $\mathbb{F}_\mathfrak{p} = \mathcal{O}_F/\mathfrak{p}$, and denote by $\mathbb{F}$ a finite field containing an isomorphic copy of each $\mathbb{F}_\mathfrak{p}$. Let $\mathbb{Q}_\mathbb{F}$ denote the fraction field of $W(\mathbb{F})$, and fix embeddings $\iota_\mathbb{F} : \mathbb{Q}_\mathbb{F} \hookrightarrow \mathbb{Q}_p^{\mathrm{ur}} \subset \overline{\mathbb{Q}}_p$, as well as embeddings $\iota_p : \overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_p$ and $\iota_\infty : \overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$. Let $\mathbb{A}_F^\infty$ denote the finite adeles over F, and $\mathbb{A}_F^{\infty,p}$ the finite prime-to-p adeles of F.

Let $\mathbb{B} = \mathrm{Emb}(F, \mathbb{Q}_\mathbb{F})$. We write $\mathbb{B} = \coprod_{\mathfrak{p}|p} \mathbb{B}_\mathfrak{p}$, where $\mathbb{B}_\mathfrak{p} = \{\beta \in \mathbb{B} \colon \mathfrak{p} \subseteq \beta^{-1}(pW(\mathbb{F}))\}$. Using ι_p, ι_∞, we identify $\mathbb{B}$ with the set of infinite places of F, and think of $\mathbb{B}_\mathfrak{p}$ as those inducing the prime ideal $\mathfrak{p}$. Let σ denote the Frobenius automorphism of $\mathbb{Q}_\mathbb{F}$, lifting $x \mapsto x^p$ modulo p. It acts on $\mathbb{B}$ by $\beta \mapsto \sigma \circ \beta$. It acts transitively on each $\mathbb{B}_\mathfrak{p}$.

Let K be a small enough (see [5, Lemma 2.4.1] for example) open compact subgroup of $\mathrm{GL}_2(\mathbb{A}_F^\infty)$ containing $\mathrm{GL}_2(\mathcal{O}_{F,p})$. Write $K = K^p K_p$, where $K^p \subset \mathrm{GL}_2(\mathbb{A}_F^{\infty,p})$, and $K_p = \mathrm{GL}_2(\mathcal{O}_{F,p})$. Fix an element $\epsilon \in (\mathbb{A}_F^{\infty,p})^\times / \det(K^p)(\widehat{\mathbb{Z}}^{(p)})^\times$, and let I denote the (prime-to-p) fractional ideal of F generated by ϵ.

We denote by X_K^ϵ the scheme of relative dimension d over $\mathrm{Spec}(\mathbb{Z}_{(p)})$ which represents the functor:

$$\mathcal{M} : \langle \mathrm{Schemes}/\mathbb{Z}_{(p)} \rangle \to \langle \mathrm{Sets} \rangle,$$

where $\mathcal{M}(S)$ is the set of all isomorphism classes of $\underline{A}/S = (A/S, \iota, \lambda, \alpha)$ such that:

- A is an abelian scheme of relative dimension d over S;
- $\iota\colon \mathcal{O}_F \hookrightarrow \mathrm{End}_S(A)$ is a ring homomorphism such that $\mathrm{Lie}(A/S)$ is locally free of rank one over $\mathcal{O}_F \otimes \mathcal{O}_S$;
- λ is a quasi-polarization inducing an $\mathcal{O}_F$-linear isomorphism $A \otimes_{\mathcal{O}_F} \mathfrak{d}_F^{-1} \xrightarrow{\sim} A^\vee \otimes_{\mathcal{O}_F} I$ (in particular, ι is invariant under the associated Rosati involution);
- α is a level K^p-structure compatible with ϵ (more precisely, in the sense of §2.1.2 of [4], with compatibility meaning that the diagram on p.12 of *loc. cit.* commutes for some ζ).

Remark 2.1. There are two standard notions of Hilbert modular varieties: Shimura varieties associated to the group $\mathrm{Res}_{F/\mathbb{Q}}\mathrm{GL}_2$, or to the group $G^* := \mathrm{Res}_{F/\mathbb{Q}}\mathrm{GL}_2 \times_{T_F} \mathbb{G}_m$, where $T_F = \mathrm{Res}_{F/\mathbb{Q}}\mathbb{G}_m$ (the morphisms to it being induced by the determinant and base-change). The advantage of the latter setting is its representability by PEL moduli problems, whereas the former is more readily applicable in the context of Langlands reciprocity. Our scheme X_K^ϵ is a model for the Shimura variety of level $K^* = g^{-1}Kg \cap G^*(\mathbb{A}_F^\infty)$ for the group G^*, where $g \in \mathrm{GL}_2(\mathbb{A}_F^{\infty,p})$ is such that $\det(g) = \epsilon$, with conventions chosen as in [4] to facilitate the transfer of results (in §10 below) to the setting of $\mathrm{Res}_{F/\mathbb{Q}}\mathrm{GL}_2$. However, we differ slightly from the set-up in [4] by imposing the compatibility condition with a fixed ϵ, as it will be more convenient here to work with (not just locally) Noetherian schemes.

We let $X = X_K^\epsilon \otimes_{\mathbb{Z}_{(p)}} \mathbb{F}$. Let $s : \mathcal{A} \to X$ denote the universal abelian scheme. Let $\omega = s_*\Omega^1_{\mathcal{A}/X}$ and $\mathcal{H} = s_*H^1_{\mathrm{dR}}(\mathcal{A}/X)$. By assumption, ω is locally free of rank one over $\mathcal{O}_X \otimes_{\mathbb{Z}} \mathcal{O}_F \cong \oplus_{\beta\in\mathbb{B}}\, \mathcal{O}_X$. This induces a decomposition

$$\omega = \oplus_{\beta\in\mathbb{B}}\, \omega_\beta,$$

where, each ω_β is locally free of rank one over $\mathcal{O}_X$. Similarly, we have a decomposition

$$\mathcal{H} = \oplus_{\beta\in\mathbb{B}}\, \mathcal{H}_\beta,$$

where, each $\mathcal{H}_\beta$ is locally free of rank two over $\mathcal{O}_X$. Furthermore, using the quasi-polarization λ and the canonical isomorphism $R^1\epsilon_*\mathcal{O}_{\mathcal{A}^\vee} \cong \mathrm{Lie}(\mathcal{A}/X)$, we obtain isomorphisms $\epsilon_\beta : \wedge^2\mathcal{H}_\beta \cong \mathcal{O}_X$.

For $\beta \in \mathbb{B}$, we let $e_\beta \in \mathbb{Z}^{\mathbb{B}}$ correspond to β. For any $\kappa = \sum k_\beta e_\beta \in \mathbb{Z}^{\mathbb{B}}$, we define

$$\omega^\kappa = \otimes_{\beta\in\mathbb{B}}\, \omega_\beta^{k_\beta}.$$

For each $\beta \in \mathbb{B}$, there is a partial Hasse invariant $\mathbf{h}_\beta \in H^0(X, \omega^{\otimes p}_{\sigma^{-1}\beta} \otimes \omega_\beta^{-1})$ defined in [1], as the β-component of the Verschiebung map $V : \omega \to \omega^{(p)}$.

The Goren-Oort stratification on X is a collection $\{X_T\}_{T\subseteq\mathbb{B}}$ of closed subschemes of X, where X_T, given by the vanishing of $\{\mathbf{h}_\beta : \beta \in T\}$, is a smooth and equi-dimensional closed subscheme of codimension $|T|$. If $\beta \notin T$, we denote the restriction of $\mathbf{h}_\beta$ to X_T by $\mathbf{h}_\beta(T)$.

Definition 2.2. Let $\mathfrak{p}|p$, and $\beta \in \mathbb{B}_\mathfrak{p}$. Assume $\beta \in T \subset \mathbb{B}$. Consider the Verschiebung morphism $V_\beta : \mathcal{H}_\beta \to \mathcal{H}^{(p)}_{\sigma^{-1}\beta}$ restricted to X_T. It is surjective onto $\omega^{(p)}_{\sigma^{-1}\beta}$, and since $\beta \in T$, it vanishes on ω_β, inducing an isomorphism of line bundles on X_T:

$$V_\beta : \omega_\beta^{-1} \overset{\epsilon_\beta}{\cong} \omega_\beta^{-1} \otimes \wedge^2\mathcal{H}_\beta \cong \mathcal{H}_\beta/\omega_\beta \xrightarrow{\sim} \omega^p_{\sigma^{-1}\beta}.$$

We let $\mathbf{b}_\beta(T)$ denote the corresponding nowhere-vanishing section in $H^0(X_T, \omega^{\otimes p}_{\sigma^{-1}\beta} \otimes \omega_\beta)$.

Corollary 2.3. *Let $\mathbb{B}_{\mathfrak{p}} \subset T \subset \mathbb{B}$. Then, for each $\beta \in \mathbb{B}_{\mathfrak{p}}$, we have $\omega_\beta^{(-p)^{|\mathbb{B}_{\mathfrak{p}}|}-1} \cong \mathcal{O}_{X_T}$.*

3. Morphisms between strata

We start by reviewing some results from [8]. Let $T = \sqcup_{\mathfrak{p}|p} T_{\mathfrak{p}}$, where $T_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$. A chain in $T_{\mathfrak{p}}$ is a subset of the form $C = \{\sigma^{-i}\beta : 0 \leq i \leq m\}$ where $m \geq 0$, and so that $\sigma^{-m-1}\beta \notin T_{\mathfrak{p}}$ and $\sigma\beta \notin T_{\mathfrak{p}}$. For such a chain, we define $\tilde{C} = C$ if m is odd, and $\tilde{C} = C \cup \{\sigma^{-m-1}\beta\}$ if m is even. Any $T_{\mathfrak{p}} \subsetneq \mathbb{B}_{\mathfrak{p}}$ can be written as a disjoint union of chains, $T_{\mathfrak{p}} = \sqcup_j C_j$, and we define $\tilde{T}_{\mathfrak{p}} = \sqcup_j \tilde{C}_j$. If $T_{\mathfrak{p}} = \mathbb{B}_{\mathfrak{p}}$, we define $\tilde{T}_{\mathfrak{p}} = \mathbb{B}_{\mathfrak{p}}$. Set $\tilde{T} = \sqcup_{\mathfrak{p}|p} \tilde{T}_{\mathfrak{p}}$, and

$$S(T) = \tilde{T} \cup \{\mathfrak{p} : T_{\mathfrak{p}} = \mathbb{B}_{\mathfrak{p}}, 2 \nmid |\mathbb{B}_{\mathfrak{p}}|\}.$$

Then $S(T)$ is a set of places of F of even cardinality, and we denote by $B_{S(T)}$ the quaternion algebra over F ramified exactly at S. We define

$$\mathrm{Iw}(T) = \{\mathfrak{p}|p : T_{\mathfrak{p}} = \mathbb{B}_{\mathfrak{p}}, \text{ and } |\mathbb{B}_{\mathfrak{p}}| \text{ is even}\}.$$

Let G denote the algebraic group over $\mathbb{Q}$ defined by

$$G(R) = \{\, g \in (B_{S(T)} \otimes R)^\times \mid \nu(g) \in R^\times \,\},$$

where ν is the reduced norm, and consider the system of quaternionic Shimura varieties $\{X_{G,K_T}\}_{K_T}$ for G, where $K_T \subset G(\mathbb{A}^\infty)$ is a compact open subgroup. Fix $T \subseteq \mathbb{B}$, and consider all T' satisfying $T \subseteq T' \subseteq \tilde{T}$, such that $\mathrm{Iw}(T') = \mathrm{Iw}(T)$. Note that $S(T) = S(T')$. It follows from Theorem 5.2 of [8] that there is a choice of K_T (see §10 below), a collection of rank 2 vector bundles $\{\mathcal{V}_\beta : \beta \in \tilde{T} - T\}$ on X_{G,K_T}, and isomorphisms

$$X_{T'} \xrightarrow[\cong]{\iota_{T'}} \Pi_{\beta \in \tilde{T}-T'} \mathbb{P}(\mathcal{V}_\beta) \xrightarrow{\pi_{T'}} X_{G,K_T},$$

where $\mathbb{P}(\mathcal{H}_\beta) \to X_{G,K_T}$ is the projectivization of $\mathcal{V}_\beta$ on X_{G,K_T}. Furthermore it follows from the proof of [8, Thm. 5.2] that there is a collection of line bundles $\{\eta_\beta : \beta \in \mathbb{B} - \tilde{T}\}$ on X_{G,K_T} such that $\iota_{T'}^* \pi_{T'}^* \eta_\beta \cong \omega_\beta$ if $\beta \in \mathbb{B} - \tilde{T}$, and $\iota_{T'}^* \mathcal{O}_\beta(-1) \cong \omega_\beta$ for any $\beta \in \tilde{T} - T'$.

We now define certain morphisms between the strata in X. Let $T, T' \subset \mathbb{B}$ be as above. Then, we have isomorphisms

$$\iota_T : X_T \simeq \Pi_{\beta \in \tilde{T}-T} \mathbb{P}(\mathcal{V}_\beta),$$
$$\iota_{T'} : X_{T'} \simeq \Pi_{\beta \in \tilde{T}-T'} \mathbb{P}(\mathcal{V}_\beta).$$

Let $\mathcal{H}_\beta^{T'} = \iota_{T'}^* \pi_{T'}^* (\mathcal{V}_\beta)$, a rank 2 vector bundle on $X_{T'}$. We obtain an isomorphism

$$\iota_{T,T'} : X_T \simeq \Pi_{\beta \in T'-T} \mathbb{P}(\mathcal{H}_\beta^{T'}).$$

Let $s_{T'}$ denote the structural morphism $\mathbb{P}(\mathcal{H}_\beta^{T'}) \to X_{T'}$. It follows that $\iota_{T,T'}^* s_{T'}^* \omega_\beta \cong \omega_\beta$ for all $\beta \in \mathbb{B} - T'$, and $\iota_{T,T'}^* \mathcal{O}_\beta(-1) \cong \omega_\beta$ for $\beta \in T' - T$. We define

$$\pi_{T,T'} := s_{T'} \circ \iota_{T,T'} : X_T \to X_{T'}.$$

Then, $\pi_{T,T'}^* \omega_\beta \cong \omega_\beta$ for $\beta \in \mathbb{B} - T'$, and $\omega_\beta \cong \mathcal{O}_\beta(-1)$ over X_T, for $\beta \in T' - T$. If $T \subset T'' \subset T'$, then $\pi_{T,T''}$ and $\pi_{T',T''}$ are defined, and

$$\pi_{T'',T'} \pi_{T,T''} = \pi_{T,T'}. \tag{3.1}$$

In the following definition, given T, we attach various integers to $\beta \in \mathbb{B}$ that we will use throughout the article.

Definition 3.1. Fix $T \subseteq \mathbb{B}$, and let $\beta \in \mathbb{B}_{\mathfrak{p}} \subseteq \mathbb{B}$. If $T_{\mathfrak{p}} \neq \mathbb{B}_{\mathfrak{p}}$, we define μ_β to be the smallest integer $i > 0$ such that $\sigma^i\beta \notin T$. If $T_{\mathfrak{p}} = \mathbb{B}_{\mathfrak{p}}$, we define $\mu_\beta = 0$. If $\beta \in \mathbb{B}_{\mathfrak{p}}$, and $\mathbb{B}_{\mathfrak{p}} - \tilde{T}_{\mathfrak{p}} \neq \emptyset$, we define n_β to be the smallest integer $i > 0$ such that $\sigma^i\beta \in \mathbb{B}_{\mathfrak{p}} - \tilde{T}_{\mathfrak{p}}$. Finally, if $\beta \in \mathbb{B}_{\mathfrak{p}} \neq T_{\mathfrak{p}}$, we define ν_β to be the smallest integer $i \geq 0$ such that $\sigma^{-i}\beta \notin T_{\mathfrak{p}}$. Note that all these notions depend on T.

Lemma 3.2. *Let $T \subseteq T' \subseteq \mathbb{B}$ be as above. Assume $\kappa = \sum_{\beta\in\mathbb{B}} k_\beta e_\beta \in \mathbb{Z}^{\mathbb{B}}$ be such that $k_{\sigma^i\beta} = 0$ for all $\beta \in T' - T, 0 \leq i \leq \mu_\beta - 1$. Then, $\pi^*_{T,T'}\omega^\kappa \cong \omega^\kappa$, and the pullback morphism*

$$\pi^*_{T,T'} : H^0(X_{T'}, \omega^\kappa) \to H^0(X_T, \omega^\kappa)$$

is an isomorphism. Furthermore, these isomorphisms are compatible in the sense that if T'' is such that $T \subseteq T'' \subseteq T' \subseteq \mathbb{B}$, then $\pi^*_{T,T''} \circ \pi^*_{T'',T'} = \pi^*_{T,T'}$.

Proof. We claim that $\pi^*_{T,T'}\omega_\beta \cong \omega_\beta$, for all $\beta \in J = \mathbb{B} - \{\sigma^i\tau : \tau \in T' - T, 0 \leq i \leq \mu_\tau - 1\}$. From the above discussion, we know this holds for all $\tau \in \mathbb{B} - T'$. Any $\beta \in J$ is of the form $\sigma^i\beta_0$, where $\beta_0 \in \mathbb{B} - T'$, and $0 \leq i \leq \mu_{\beta_0} - 1$. The claim now follows since for such β_0, i, we have an isomorphism $\omega_{\sigma^i\beta_0} \cong \omega_{\beta_0}^{(-p)^i}$, over both X_T and $X_{T'}$. The main statement now follows since $\pi_{T,T'} : X_T \to X_{T'}$ is a product of $\mathbb{P}^1$-bundles. The compatibility assertion follows from Equation (3.1) and the definition of the isomorphism $\pi^*_{T,T'}\omega^\kappa \cong \omega^\kappa$. □

4. Cones of weight for strata

In the following we will prove the existence of a collection of sections of ω^κ on a stratum X_T (for varying κ) compatibly with respect to the morphisms described above.

Let $\mathfrak{p}|p$, and $\beta, \beta' \in \mathbb{B}_{\mathfrak{p}}$. Choose $0 < n \leq |\mathbb{B}_{\mathfrak{p}}|$ such that $\sigma^n\beta' = \beta$. We define $h_\beta^{\beta'} := -e_\beta + p^n e_{\beta'}$, and $b_\beta^{\beta'} := e_\beta + p^n e_{\beta'}$. When $n = 1$, we simply write h_β, and b_β, so that we have $\mathbf{h}_\beta(T) \in H^0(X_T, \omega^{h_\beta})$, for $\beta \notin T$, and $\mathbf{b}_\beta(T) \in H^0(X_T, \omega^{b_\beta})$, for $\beta \in T$. We have an identity:

$$h^{\sigma^n\beta}_{\sigma^{n+m}\beta} + p^m h^{\beta}_{\sigma^n\beta} = h^{\beta}_{\sigma^{n+m}\beta}, \tag{4.1}$$

if $n, m > 0$, and $n + m \leq |\mathbb{B}_{\mathfrak{p}}|$.

For each $T \subseteq \mathbb{B}$, and $\mathfrak{p}|p$, let $\mathcal{H}(T_{\mathfrak{p}}) = \mathbb{B}_{\mathfrak{p}} - T_{\mathfrak{p}}$, and $\mathcal{T}(T_{\mathfrak{p}}) = (\mathbb{B}_{\mathfrak{p}} - \tilde{T}_{\mathfrak{p}}) \cup \sigma(\tilde{T}_{\mathfrak{p}} - T_{\mathfrak{p}})$.

Proposition 4.1. *Let $\beta \in \mathcal{H}(T_{\mathfrak{p}})$, and $\beta' \in \mathcal{T}(T_{\mathfrak{p}})$. Write $\beta' = \sigma^{-m}\beta$, where $0 < m \leq |\mathbb{B}_{\mathfrak{p}}|$. Then, there exists a nonzero section*

$$\mathbf{h}_\beta^{\beta'}(T) \in H^0(X_T, \omega^{h_\beta^{\beta'}}),$$

such that if $T \subseteq T' \subseteq T^0 := T \cup \big(\{\sigma^{-j}\beta : 0 < j < m\} \cap \tilde{T}\big)$ (whence, $\beta \in \mathcal{H}(T'_{\mathfrak{p}}), \beta' \in \mathcal{T}(T'_{\mathfrak{p}})$, and $\pi_{T,T'}$ is defined), then

$$\mathbf{h}_\beta^{\beta'}(T) = \pi^*_{T,T'}\mathbf{h}_\beta^{\beta'}(T').$$

Proof. For $\tau \in \mathbb{B}_{\mathfrak{p}}$, let m_τ be the smallest $j \geq 0$ such that $\sigma^j\tau \notin T^0$, and for $0 \leq i \leq m$, set $\delta_i = (-1)^{m_{\sigma^{-i}\beta}}$. Set $\mathbf{f}_\tau = \mathbf{b}_\tau(T^0)$ if $\tau \in T^0$, and $\mathbf{f}_\tau = \mathbf{h}_\tau(T^0)$ otherwise. Define

$$\mathbf{h}_\beta^{\beta'}(T) = \pi^*_{T,T^0} \prod_{0\leq i<m} (\mathbf{f}_{\sigma^{-i}\beta})^{\delta_i p^i},$$

where $\prod_{0\leq i<m}(\mathbf{f}_{\sigma^{-i}\beta})^{\delta_i p^i}$ is a (nonzero) section of $\omega^{h_\beta^{\beta'}}$ on X_{T^0}, noting that the exponent $\delta_i p^i$ is negative only if $\sigma^{-i}\beta \in T_0$, implying $\mathbf{f}_{\sigma^{-i}\beta}^{-1} = \mathbf{b}_{\sigma^{-i}\beta}(T^0)^{-1}$ is defined on X_{T^0}. By Lemma 3.2, it follows that $\mathbf{h}_\beta^{\beta'}(T)$ is a nonzero section of $\omega^{h_\beta^{\beta'}}$ on X_T. The second statement then follows from the definition of the sections and the compatibility assertion in the lemma. □

Lemma 4.2. *Let $T \subseteq \mathbb{B}$, and $\mathfrak{p}|p$. Let $\beta \in \mathcal{H}(T_\mathfrak{p})$, and $\beta' \in \mathcal{T}(T_\mathfrak{p})$. Then the vanishing locus of $\mathbf{h}_\beta^{\beta'}(T)$ in X_T intersects every connected component of X_T in a nonempty smooth divisor.*

Proof. First note that if $\iota : Y_1 \to Y_2$ is a product of $\mathbb{P}^1$- bundles, and f a section on Y_2, the vanishing locus of which intersects every connected component of Y_2 in a nonempty smooth divisor, then the vanishing locus of $\iota^* f$ intersects every connected component of Y_1 in a nonempty smooth divisor. Hence, in the notation of Proposition 4.1, it is enough to prove the statement for $\mathbf{h}_\beta^{\beta'}(T^0)$. This follows from Lemma 4.1 of [2]. □

Let Z be a closed subscheme of X. We define $C(Z, \omega^\bullet)$ to be the cone in $\mathbb{Z}^\mathbb{B}$ generated by

$$\{\kappa \in \mathbb{B}^\mathbb{Z} : H^0(Z, \omega^\kappa) \neq 0\}.$$

Let $\mathcal{C}(Z, \omega^\bullet)$ be the $\mathbb{Q}^{\geq 0}$-saturation of $C(Z, \omega^\bullet)$. When $Z = X_T$, we denote $C(X_T, \omega^\bullet), \mathcal{C}(X_T, \omega^\bullet)$ by $C_T, \mathcal{C}_T$, respectively. We always have

$$\langle \pm b_\beta : \beta \in T; h_\beta : \beta \notin T\rangle \subset C_T,$$

appearing as weights of $\{\mathbf{h}_\beta(T) : \beta \notin T\}$ and $\{\mathbf{b}_\beta(T)^{\pm 1} : \beta \in T\}$ on X_T.

Let $T = \sqcup_{\mathfrak{p}|p} T_\mathfrak{p}$, where for each $\mathfrak{p}|p$, $T_\mathfrak{p} \subseteq \mathbb{B}_\mathfrak{p}$. Let $\tilde{T} = \sqcup_{\mathfrak{p}|p}\tilde{T}_\mathfrak{p}$ be as in §3

Definition 4.3. Let $\mathfrak{p}|p$. Recall that $\mathcal{H}(T_\mathfrak{p}) = \mathbb{B}_\mathfrak{p} - T_\mathfrak{p}$, and $\mathcal{T}(T_\mathfrak{p}) = (\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}) \cup \sigma(\tilde{T}_\mathfrak{p} - T_\mathfrak{p})$. We define

$$\mathcal{G}(T_\mathfrak{p}) = \{h_\beta^{\beta'} : \beta \in \mathcal{H}(T_\mathfrak{p}), \beta' \in \mathcal{T}(T_\mathfrak{p})\} \cup \{\pm b_\beta : \beta \in T_\mathfrak{p}\}.$$

We define $D_{T_\mathfrak{p}}$ to be the $\mathbb{Z}^{\geq 0}$-cone generated by $\mathcal{G}(T_\mathfrak{p})$, and D_T the $\mathbb{Z}^{\geq 0}$-cone generated by $D_{T_\mathfrak{p}}$ for all $\mathfrak{p}|p$. We define $\mathcal{D}_{T_\mathfrak{p}}$ and $\mathcal{D}_T$ to be the $\mathbb{Q}^{\geq 0}$-saturation of $D_{T_\mathfrak{p}}$ and D_T, respectively.

Corollary 4.4. *For any $T \subseteq \mathbb{B}$, we have $D_T \subseteq C_T$.*

Proof. It is enough to prove that $D_{T_\mathfrak{p}} \subseteq C_T$, for all $\mathfrak{p}|p$, which follows from Proposition 4.1. □

5. Reduced cones and optimal bases

We provide an optimal basis for $\mathcal{D}_T$. Fix $\mathfrak{p}|p$. First assume that $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} \neq \emptyset$. Recall n_β from Definition 3.1. We define

$$\mathcal{G}'(T_\mathfrak{p}) = \{h_{\sigma^{n_\beta}\beta}^{\beta} : \beta \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}\} \cup \{-b_{\sigma^{n_\beta}\beta}^{\beta} : \beta \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}\} \cup \{\pm b_\beta : \beta \in T_\mathfrak{p}\}.$$

When $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} = \emptyset$, we define

$$\mathcal{G}'(T_\mathfrak{p}) = \{-e_\beta : \beta \in \mathbb{B} - T_\mathfrak{p}\} \cup \{\pm b_\beta : \beta \in T_\mathfrak{p}\}.$$

In either case, we define $D'_{T_\mathfrak{p}}$ to be the $\mathbb{Z}^{\geq 0}$-cone generated by $\mathcal{G}'(T_\mathfrak{p})$ in $\mathbb{Z}^{\mathbb{B}_\mathfrak{p}}$, and $\mathcal{D}'_{T_\mathfrak{p}}$ to be the $\mathbb{Q}^{\geq 0}$-saturation of it. Let $\mathcal{D}'_T$ be the $\mathbb{Q}^{\geq 0}$-cone generated by $\{\mathcal{D}'_{T_\mathfrak{p}} : \mathfrak{p}|p\}$ in $\mathbb{Q}^\mathbb{B}$.

Proposition 5.1. *We have $\mathcal{D}'_T = \mathcal{D}_T$, for all $T \subseteq \mathbb{B}$.*

Proof. We begin by proving $\mathcal{D}'_{T_\mathfrak{p}} \subseteq \mathcal{D}_{T_\mathfrak{p}}$. First, assume $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} \neq \emptyset$. Clearly, if $\beta \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}$, then $h^{\beta}_{\sigma^{n_\beta}\beta}$ lies in $\mathcal{D}_{T_\mathfrak{p}}$. If $\beta \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}$, then

$$-b^{\beta}_{\sigma^{n_\beta}\beta} = h^{\sigma\beta}_{\sigma^{n_\beta}\beta} - p^{(n_\beta-1)} b_{\sigma\beta}.$$

Since $\sigma^{n_\beta}\beta \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}$, and $\sigma\beta \in \sigma(\tilde{T}_\mathfrak{p} - T_\mathfrak{p}) \subset T_\mathfrak{p}$, it follows that both $h^{\sigma\beta}_{\sigma^{n_\beta}\beta}$ and $-b_{\sigma\beta}$ lie in $\mathcal{D}_{T_\mathfrak{p}}$, and, hence, so does $-b^{\beta}_{\sigma^{n_\beta}\beta}$. If $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} = \emptyset$, then for $\beta \in \mathbb{B}_\mathfrak{p} - T_\mathfrak{p} = \tilde{T}_\mathfrak{p} - T_\mathfrak{p}$, we have $-(1+p^{|\mathbb{B}_\mathfrak{p}|})e_\beta = h^{\sigma\beta}_{\beta} - p^{|\mathbb{B}_\mathfrak{p}|-1} b_{\sigma\beta} \in D_{T_\mathfrak{p}}$, and hence $-e_\beta \in \mathcal{D}_{T_\mathfrak{p}}$.

We now prove the reverse inclusion. Note that for all $\beta \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}$, we have $(p^{|\mathbb{B}_\mathfrak{p}|}-1)e_\beta = h^{\beta}_{\beta} \in D'_{T_\mathfrak{p}}$, and hence $e_\beta \in \mathcal{D}'_{T_\mathfrak{p}}$. Also, for all $\beta \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}$, we have $-e_\beta \in \mathcal{D}'_{T_\mathfrak{p}}$. This is clear if $\tilde{T}_\mathfrak{p} = \mathbb{B}_\mathfrak{p}$. Otherwise, note that $-p^{n_\beta}e_\beta = -b^{\beta}_{\sigma^{n_\beta}\beta} + e_{\sigma^{n_\beta}\beta} \in \mathcal{D}_{T'_\mathfrak{p}}$, as $\sigma^{n_\beta}\beta \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}$, whence $e_{\sigma^{n_\beta}\beta} \in \mathcal{D}'_{T_\mathfrak{p}}$

To prove $\mathcal{D}_{T_\mathfrak{p}} \subset \mathcal{D}'_{T_\mathfrak{p}}$, we consider $h^{\beta'}_{\beta}$ in four cases:

Case 1: $\beta, \beta' \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}$. By Identity 4.1, we have $\langle h^{\beta}_{\sigma^{n_\beta}\beta} : \beta \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}\rangle = \langle h^{\beta'}_{\beta} : \beta, \beta' \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}\rangle$. Hence $h^{\beta'}_{\beta} \in D'_{T_\mathfrak{p}}$ by definition.

Case 2: $\beta \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}, \beta' \in \sigma(\tilde{T}_\mathfrak{p} - T_\mathfrak{p}) \subseteq T_\mathfrak{p}$. By Identity 4.1, it is enough to assume $\beta = \sigma^{n_{\beta'}}\beta'$, in which case, $h^{\beta'}_{\beta} = -b^{\sigma^{-1}\beta'}_{\beta} + p^{n_{\beta'}} b_{\beta'} \in \mathcal{D}'_{T_\mathfrak{p}}$.

Case 3: $\beta \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}, \beta' \in \sigma(\tilde{T}_\mathfrak{p} - T_\mathfrak{p}) \subseteq T_\mathfrak{p}$. Assume $\sigma^n\beta' = \beta$. In this case, we have $h^{\beta'}_{\beta} = -e_\beta - p^{n+1}e_{\sigma^{-1}\beta'} + p^n b_{\beta'} \in \mathcal{D}'_{T_\mathfrak{p}}$, since $-e_\beta, -e_{\sigma^{-1}\beta'}, b_{\beta'} \in \mathcal{D}'_{T_\mathfrak{p}}$.

Case 4: $\beta \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}, \beta' \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}$. Assume $\sigma^n\beta' = \beta$. We have $h^{\beta'}_{\beta} = -e_\beta + p^n e_{\beta'} \in \mathcal{D}'_{T_\mathfrak{p}}$, since $-e_\beta, e_{\beta'} \in \mathcal{D}'_{T_\mathfrak{p}}$. □

Lemma 5.2. *When $\tilde{T} = T$, we have $\mathcal{D}_T = \langle h_\beta : \beta \in \mathbb{B} - T\rangle + \langle \pm b_\beta : \beta \in T\rangle$.*

Proof. By Proposition 5.1, we have $\mathcal{D}_T = \mathcal{D}'_T = \langle h^{\beta}_{\sigma^{n_\beta}\beta} : \beta \in \mathbb{B} - T\rangle + \langle \pm b_\beta : \beta \in T\rangle$. The result follows since for $\beta \notin T = \tilde{T}$, we have $h^{\beta}_{\sigma^{n_\beta}\beta} = h_{\sigma^{n_\beta}\beta} + \Sigma_{i=1}^{n_\beta - 1}(-p)^{n_\beta - i} b_{\sigma^i\beta}$, noting that each n_β is odd in this case. □

For any $\mathbb{Q}^{\geq 0}$-cone $\langle \pm b_\beta : \beta \in T\rangle \subset C \subset \mathbb{Q}^{\mathbb{B}}$, let $C^{T-\mathrm{red}}$ be the image of C under

$$i_T : \mathbb{Q}^{\mathbb{B}} \to \mathbb{Q}^{\mathbb{B}-T},$$

which is defined as follows: for $\beta \in \mathbb{B}_\mathfrak{p} \subseteq \mathbb{B}$, we have $\pi(e_\beta) = 0$ if $\mathbb{B}_\mathfrak{p} = T_\mathfrak{p}$, and, otherwise, $\pi(e_\beta) = (-p)^{\nu_\beta} e_{\sigma^{-\nu_\beta}\beta}$, where ν_β is given in Definition 3.1. Hence,

$$i_T(\kappa) = \sum_{\beta \in \mathbb{B}-T} \Big(\sum_{i=0}^{\mu_\beta - 1} (-p)^i k_{\sigma^i\beta} \Big) e_\beta. \tag{5.1}$$

The kernel of π is the cone generated by all $\{\pm b_\beta : \beta \in T_\mathfrak{p}\}$, and the obvious inclusion $j_T : C^{T-\mathrm{red}} \to C$ gives a section to π. Hence, we have $C = j_T(C^{T-\mathrm{red}}) + \langle \pm b_\beta : \beta \in T\rangle$. It follows that $D_T^{T-\mathrm{red}}$ is generated by $\mathcal{G}^{T-\mathrm{red}}(T) = \cup_{\mathfrak{p}|p} \mathcal{G}_\mathfrak{p}^{T-\mathrm{red}}(T)$, where,

$$\mathcal{G}_\mathfrak{p}^{T-\mathrm{red}}(T) = \{h^{\beta}_{\sigma^{n_\beta}\beta} : \beta \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}\} \cup \{-b^{\beta}_{\sigma^{n_\beta}\beta} : \beta \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}\},$$

if $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} \neq \emptyset$, and $\mathcal{G}_\mathfrak{p}^{T-\mathrm{red}}(T) = \{-e_\beta : \beta \in \mathbb{B} - T_\mathfrak{p}\}$, if $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} = \emptyset$.

6. Explicit description of $\mathcal{D}_T$

We will now obtain an explicit description of $\mathcal{D}_T$ in terms of the weight components, i.e., we determine a basis for the dual cone of $\mathcal{D}_T^{T-\mathrm{red}}$. For $\beta \in \mathbb{B} - \tilde{T}$, we set $g_\beta = h^\beta_{\sigma^{n_\beta}\beta}$, if $\beta \in \mathbb{B} - \tilde{T}$, and $g_\beta = b_\beta$ if $\beta \in T$. For $\beta \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}$, we set $g_\beta = -b^\beta_{\sigma^{n_\beta}\beta}$, if $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} \neq \emptyset$, and $g_\beta = -e_\beta$, otherwise. By Proposition 5.1, we have

$$\mathcal{D}_T = \langle g_\beta : \beta \in \mathbb{B} - T\rangle + \langle \pm g_\beta : \beta \in T\rangle.$$

Let $\epsilon : \mathbb{B} \to \{-1, 0, 1\}$ be a map. Let $T \subseteq \mathbb{B}$, $\mathfrak{p}|p$, and $\beta, \beta' \in \mathbb{B}_\mathfrak{p}$. Write $\beta' = \sigma^n\beta$ for some $0 \leq n < |\mathbb{B}_\mathfrak{p}|$. define the functional

$$L^\epsilon[\beta, \beta'](\kappa) = \sum_{0\leq i\leq n} \epsilon(\sigma^i\beta)p^i k_{\sigma^i\beta}.$$

We simply write $L^\epsilon[\beta]$ for $L^\epsilon[\beta, \sigma^{-1}\beta]$.

Given $T \subset \mathbb{B}$, we define its sign function $\epsilon_T : \mathbb{B} \to \{-1, 1\}$. If $\beta \in \mathbb{B}_\mathfrak{p}$ and $T_\mathfrak{p} = \mathbb{B}_\mathfrak{p}$, we define $\epsilon_T(\beta) = 0$. Otherwise, we define $\epsilon_T(\beta) = 1$ for $\beta \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}$, and $\epsilon_T(\beta) = (-1)^{\mu_\beta - 1}$ for $\beta \in \tilde{T}_\mathfrak{p}$. We define for $\beta \in \mathbb{B} - T$:

$$L_{T,\beta} = \begin{cases} L^{\epsilon_T}[\sigma^{\mu_\beta}\beta] & \text{if } \beta \in \mathbb{B} - \tilde{T}, \\ L^{\epsilon_T}[\beta, \sigma^{\mu_\beta - 1}\beta] & \text{if } \beta \in \tilde{T} - T. \end{cases}$$

Proposition 6.1. *Let $T \subset \mathbb{B}$. We have*

$$\mathcal{D}_T = \{\kappa \in \mathbb{Q}^\mathbb{B} : L_{T,\beta}(\kappa) \geq 0,\ \forall \beta \in \mathbb{B} - T\}.$$

Proof. Note that $\{g_\beta : \beta \in \mathbb{B}\}$ is a basis for $\mathbb{Q}^\mathbb{B}$. By inspection, for $\beta \in \mathbb{B} - T$, we have

$$L_{T,\beta}(g_\tau) = \begin{cases} > 0 & \text{if } \tau = \beta \\ = 0 & \text{if } \tau \neq \beta. \end{cases}$$

Since $\mathcal{D}_T = \langle g_\beta : \beta \in \mathbb{B} - T\rangle + \langle \pm g_\beta : \beta \in T\rangle$, it follows that

$$\mathcal{D}_T = \{\kappa \in \mathbb{Q}^\mathbb{B} : L_{T,\beta}(\kappa) \geq 0,\ \forall \beta \in \mathbb{B} - T\}.$$

□

7. The Main Theorem

In this section, we determine the cones of weights of strata of X. First, we prove a lemma. Let Y be a scheme over $\mathbb{F}$, and $\{\mathcal{L}_i\}_{i\in I}$ a finite collection of line bundles on Y. For any weight $\kappa = \sum_{i\in I} \kappa_i e_i \in \mathbb{Z}^I$, we define $\mathcal{L}^\kappa = \otimes_{i\in I}\mathcal{L}_i^{\kappa_i}$. If Z is a closed subscheme of Y, we define $C_Z(\mathcal{L}^\bullet)$ to be the $\mathbb{Q}^{\geq 0}$-cone in $\mathbb{Z}^I$ generated by the following subset of $\mathbb{Z}^I$:

$$\{\kappa \in \mathbb{Z}^I : H^0(Z, \mathcal{L}^\kappa) \neq 0\}.$$

Lemma 7.1. *Let Y be a noetherian scheme over $\mathbb{F}$, and $\{\mathcal{L}_i\}_{i\in I}$ a finite collection of line bundles on Y. For $j \in J$ (a finite set), let $\mathbf{h}_j \in H^0(Y, \mathcal{L}^{h_j})$, where $h_j \in \mathbb{Z}^I$. Let Z_j be the vanishing locus of $\mathbf{h}_j$, and assume that each Z_j is purely of codimension one. Assume further that the intersection of Z_j with every irreducible component of Y is nonempty. Then,*

$$C_Y(\mathcal{L}^\bullet) \subset \big(\bigcap_{j\in J} C_{Z_j}(\mathcal{L}^\bullet)\big) + \langle h_j : j \in J\rangle,$$

where $\langle h_j : j \in J\rangle$ denotes the $\mathbb{Q}^{\geq 0}$-cone generated by $\{h_j : j \in J\}$.

Proof. Assume $f \neq 0$ is a section of $\mathcal{L}^\kappa$ on Y. By Lemma 7.2 in [3], we can write $f = (\Pi_{j\in J}\mathbf{h}_j^{M_j})f_0$ for some nonnegative integers $\{M_j : j \in J\}$, such that restriction of f_0 to each Z_j is nonzero. Therefore, f_0 gives a nonzero section of $\mathcal{L}^{\otimes(\kappa-\sum_{j\in J}M_jh_j)}$ on each Z_j. It follows that $\kappa - \sum_{j\in J}M_jh_j \in \bigcap_{j\in J}C_{Z_j}(\mathcal{L}^\bullet)$. □

Remark 7.2. Expressions involving cones of weights such as the one appearing in the above lemma arise, and are referred to as intersection-sum cones, in the work of Goldring-Koskivirta on weight cone conjectures. They also implicitly appear in the work of Diamond-Kassaei on minimal cones.

Proposition 7.3. *Let $T \subseteq \mathbb{B}$. Let $T \subseteq T' \subseteq \tilde{T}$ be such that $\mathrm{Iw}(T') = \mathrm{Iw}(T)$ (so, in particular, the morphism $\pi_{T,T'} : X_T \to X_{T'}$ is defined). Let Z be a reduced closed subscheme of $X_{T'}$. Then*

$$\mathcal{C}(\pi_{T,T'}^{-1}Z, \omega^\bullet) \subset \{\kappa = \sum_{\beta\in\mathbb{B}} k_\beta e_\beta : L_{T,\beta_0}(\kappa) \geq 0, \forall \beta_0 \in T' - T\}.$$

Proof. Note that if $T_\mathfrak{p} = \mathbb{B}_\mathfrak{p}$, for some $\mathfrak{p}|p$, there is nothing to prove at $\beta_0 \in \mathbb{B}_\mathfrak{p}$. Furthermore, by Corollary 2.3, it is true that for each $\beta \in \mathbb{B}_\mathfrak{p}$, the line bundle ω_β is torsion on X_T. Fix $\mathfrak{p}|p$ such that $T_\mathfrak{p} \neq \mathbb{B}_\mathfrak{p}$. For any $\beta \in \mathbb{B}_\mathfrak{p}$, define ν_β to be the smallest $i \geq 0$ such that $\sigma^{-i}\beta \notin T_\mathfrak{p}$. Then, from Definition 2.2, we have $\omega_\beta \cong \omega_{\sigma^{-\nu_\beta}\beta}^{(-p)^{\nu_\beta}}$. For any $\beta \in T_\mathfrak{p}$, the embedding $\sigma^{-v_\beta}\beta$ is in $\mathbb{B}_\mathfrak{p} - T_\mathfrak{p}$, so it is either in $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}$, or in $\tilde{T}_\mathfrak{p} - T_\mathfrak{p}$. Putting all this together, and noting that the subset of $\beta \in \mathbb{B}_\mathfrak{p}$ for which $\sigma^{-\nu_\beta}\beta = \beta_0 \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}$ is exactly $\{\beta_0, \sigma\beta_0, ..., \sigma^{\mu_{\beta_0}-1}\beta_0\}$, we find that there are integers $\ell_\beta(\kappa)$, for $\beta \in \mathbb{B} - \tilde{T}$, such that for some $N > 0$, we have an isomorphism of line bundles on X_T:

$$\omega^{N\kappa} \cong \otimes_{\beta\in\mathbb{B}-\tilde{T}}\omega_\beta^{N\ell_\beta(\kappa)} \otimes \otimes_{\beta\in\tilde{T}-T}\omega_\beta^{NL_{T,\beta}(\kappa)}.$$

By assumptions, we have an isomorphism $\iota_{T,T'} : X_T \to \Pi_{\beta\in I}\mathbb{P}(\mathcal{H}_\beta^{T'})$ (see §3). Under $\iota_{T,T'}$, this line bundle corresponds to $\mathcal{L}_N = \otimes_{\beta\in\mathbb{B}-\tilde{T}}s_{T'}^*\omega_\beta^{N\ell_\beta(\kappa)} \otimes \otimes_{\beta\in\tilde{T}-T'}s_{T'}^*\omega_\beta^{NL_{T,\beta}(\kappa)} \otimes \otimes_{\beta\in T'-T}\mathcal{O}_\beta(NL_{T,\beta}(\kappa))$.

Now assume that κ is a weight such that $L_{T,\beta_0}(\kappa) < 0$, for some $\beta_0 \in T' - T$, and let $f \in H^0(\pi_{T,T'}^{-1}Z, \omega^\kappa)$. Then, under the isomorphism $i_{T,T'}|_{\pi_{T,T'}^{-1}Z} : \pi_{T,T'}^{-1}Z \to \Pi_{\beta\in T'-T}\mathbb{P}(\mathcal{H}_\beta^{T'}|_Z)$, the section f^N of $\omega^{N\kappa}$ on $\pi_{T,T'}^{-1}Z$ gives rise to a section of $\mathcal{L}_N$ on $\Pi_{\beta\in T'-T}\mathbb{P}(\mathcal{H}_\beta^{T'}|_Z)$. The restriction of $\mathcal{L}_N$ to every geometric fibre of $s_{T'} : \Pi_{\beta\in T'-T}\mathbb{P}(\mathcal{H}_\beta^{T'}|_Z) \to Z$ is isomorphic to $\otimes_{\beta\in T'-T}\mathcal{O}_\beta(NL_{T,\beta}(\kappa))$. Since $L_{T,\beta_0}(\kappa) < 0$, the restriction of f^N to any such fibre is zero. Since Z is reduced, so is $\pi_{T,T'}^{-1}Z$, and it follows that $f^N = 0$, and hence $f = 0$. □

Lemma 7.4. *Let $T \subseteq \mathbb{B}$ be such for some $\mathfrak{p}|p$, we have $|\mathbb{B}_\mathfrak{p}|$ is even, and $T_\mathfrak{p} = \mathbb{B}_\mathfrak{p} - \{\beta\}$, for some $\beta \in \mathbb{B}_\mathfrak{p}$. Then $\mathcal{C}_T \subseteq \{\kappa \in \mathbb{Q}^\mathbb{B} : L_{T,\beta}(\kappa) \geq 0\}$.*

Proof. Note that the assumptions imply $\beta \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}$. This case does not follow from Proposition 7.3, since $\mathrm{Iw}(T) \neq \mathrm{Iw}(T\cup\{\beta\})$. We prove this by induction on $|\mathbb{B}-T|$. When $|\mathbb{B}-T| = 1$, we have $T = \mathbb{B}-\{\beta\}$, and X_T is a union of complete curves. First note that, since $\omega_{\sigma^{-1}\beta} \cong \omega_\beta^{(-p)^{|\mathbb{B}_\mathfrak{p}|-1}}$, the Hasse invariant $\mathbf{h}_\beta(T)$ gives a section of $\omega_\beta^{-1-p^{|\mathbb{B}_\mathfrak{p}|}}$ whose vanishing locus meets every connected component of X_T, implying that ω_β is an anti-ample bundle on each such component. Now let $0 \neq \mathbf{f} \in H^0(X_T, \omega^\kappa)$. Since $\mathbb{B}_\mathfrak{q} \subset T$ for all $\mathfrak{q} \neq \mathfrak{p}$, by Corollary 2.3, we know that ω_τ is torsion for all $\beta \in \mathbb{B} - \mathbb{B}_\mathfrak{p}$. Hence, there is $N > 0$ such that $\mathbf{f}^N$ gives a section of $\bigotimes_{\tau\in\mathbb{B}_\mathfrak{p}}\omega_\tau^{Nk_\tau}$ on X_T. Since

all $\beta \neq \tau \in \mathbb{B}_{\mathfrak{p}}$ belong to $T_{\mathfrak{p}}$, it follows that $\bigotimes_{\tau \in \mathbb{B}_{\mathfrak{p}}} \omega_\tau^{Nk_\tau} \cong \omega_\beta^{-NL_{T,\beta}(\kappa)}$. Since ω_β is anti-ample on each component, it follows that $L_{T,\beta}(\kappa) \geq 0$.

Now assume $|\mathbb{B} - T| > 1$, and choose $\beta' \in \mathbb{B}_{\mathfrak{q}} - T_{\mathfrak{q}}$, for some $\mathfrak{q} \neq \mathfrak{p}$. Lemma 7.1 implies

$$\mathcal{C}_T \subset \mathcal{C}(X_T \cap Z(\mathbf{h}_{\beta'}(T)), \omega^\bullet) + \langle h_{\beta'} \rangle = \mathcal{C}_{T \cup \{\beta'\}} + \langle h_{\beta'} \rangle \subset \{\kappa \in \mathbb{Q}^{\mathbb{B}} : L_{T,\beta}(\kappa) \geq 0\},$$

by the induction hypothesis, and since $L_{T,\beta}(h_{\beta'}) = 0$. □

Corollary 7.5. *Let $T \subseteq \mathbb{B}$. Then*

$$\mathcal{C}(X_T, \omega^\bullet) \subset \{\kappa = \sum_{\beta \in \mathbb{B}} k_\beta e_\beta : L_{T,\beta_0}(\kappa) \geq 0, \forall \beta_0 \in \tilde{T} - T\}.$$

Proof. Let $\kappa \in \mathcal{C}(X_T, \omega^\bullet)$, and $\mathfrak{p}|p$. If either $|\mathbb{B}_{\mathfrak{p}}|$ is odd, or $\tilde{T}_{\mathfrak{p}} \neq \mathbb{B}_{\mathfrak{p}}$ it follows from Proposition 7.3 that $L_\beta(\kappa) \geq 0$, for all $\beta \in \tilde{T}_{\mathfrak{p}} - T_{\mathfrak{p}}$. Otherwise, assume $\tilde{T}_{\mathfrak{p}} = \mathbb{B}_{\mathfrak{p}}$, and $\tilde{T}_{\mathfrak{p}} - T_{\mathfrak{p}} = \{\beta_1, ..., \beta_t\}$. If $t \geq 2$, applying Proposition 7.3 with $T' = T \cup \{\beta_i\}$, for $1 \leq i \leq t$, implies that $L_{\beta_i}(\kappa) \geq 0$. When $t = 1$, Lemma 7.4 implies that $L_{\beta_1}(\kappa) \geq 0$. □

We now prove the main result.

Theorem 7.6. *For any $T \subset \mathbb{B}$, we have*

$$\mathcal{C}_T = \mathcal{D}_T = \langle h^\beta_{\sigma^{n_\beta}\beta} : \beta \in \mathbb{B} - \tilde{T} \rangle + \langle -b^\beta_{\sigma^{n_\beta}\beta} : \beta \in \tilde{T}_{\mathfrak{p}} - T_{\mathfrak{p}} \rangle + \langle \pm b_\beta : \beta \in T \rangle.$$

Proof. By Corollary 4.4 and Proposition 5.1, it is enough to show $\mathcal{C}_T \subseteq \mathcal{D}_T$. We prove this by induction on $|\tilde{T} - T|$. The initial case where $\tilde{T} = T$ is already known and can be proved in different ways. In this case, by Proposition 5.1, and in the notation of §5, we have

$$\mathcal{D}_T = \langle h^\beta_{\sigma^{n_\beta}\beta} : \beta \in \mathbb{B} - T \rangle + \langle \pm b_\beta : \beta \in T \rangle.$$

One can now either use Lemma 5.2, and appeal to [6] (where the admissibility condition on a stratum X_T corresponds to the condition $\tilde{T} = T$), or note that the proof of Corollary 5.4 in [2] applies verbatim to show that, in the notation of §3, the cone of weights $\mathcal{C}(X_{G,K_T}, \eta^\bullet)$ is generated by the weights of the Hasse invariants of X_{G,K_T}. It then follows immediately, using $\iota_T : X_T \cong X_{G,K_T}$, that

$$\mathcal{C}_T^{T-\mathrm{red}} = \iota_T^* \mathcal{C}(X_{G,K_T}, \eta^\bullet).$$

Every Hasse invariant of X_{G,K_T} is a section of a line bundle of the form $(\eta_\beta)^{p^{n_\beta}}(\eta_{\sigma^{n_\beta}\beta})^{-1}$, where $\beta \in \mathbb{B} - T$. It pulls back under ι_T to a nonzero from of weight $h^\beta_{\sigma^{n_\beta}\beta}$. We deduce that

$$\mathcal{C}_T = j_T(\mathcal{C}_T^{T-\mathrm{red}}) + \langle \pm b_\beta : \beta \in T \rangle = \langle h^\beta_{\sigma^{n_\beta}\beta} : \beta \in \mathbb{B} - T \rangle + \langle \pm b_\beta : \beta \in T \rangle = \mathcal{D}_T.$$

We now assume $\tilde{T} \neq T$. Choose $\mathfrak{p}|p$ such that $\tilde{T}_{\mathfrak{p}} - T_{\mathfrak{p}} = \{\beta_1, ..., \beta_r\}$ for $r > 0$. We consider two cases:

Case 1: $\mathbb{B}_{\mathfrak{p}} - \tilde{T}_{\mathfrak{p}} \neq \emptyset$. Without loss of generality, we can assume there is $\beta \in \mathbb{B}_{\mathfrak{p}} - \tilde{T}_{\mathfrak{p}}$ such that $\sigma^{\mu_{\beta_1}}\beta_1 = \beta$. Let $T'' = T \cup \{\beta\}$, $T' = T \cup \{\beta_1\}$. Apply Lemma 7.1 with $\pi^*_{T,T'}\mathbf{h}^\beta_\beta(T') = \mathbf{h}^\beta_\beta(T)$ and $\mathbf{h}_\beta(T)$ (using Lemma 4.2):

$$\mathcal{C}_T \subseteq \big(\mathcal{C}_{T''} \cap \mathcal{C}(Z(\pi^*_{T,T'}\mathbf{h}^\beta_\beta(T')), \omega^\bullet)\big) + \langle h_\beta, h^\beta_\beta \rangle.$$

Applying Proposition 7.3 with $Z = Z(\mathbf{h}^\beta_\beta(T'))$, we have

$$\mathcal{C}_T \subseteq \big(\mathcal{C}_{T''} \cap \{\kappa : L_{T,\beta_1}(\kappa) \geq 0\}\big) + \langle h_\beta, h^\beta_\beta \rangle.$$

By induction hypothesis, $\mathcal{C}_{T''} = \mathcal{D}_{T''}$. Since $h_\beta, h_\beta^\beta \in \mathcal{D}_T$, to show $\mathcal{C}_T \subseteq \mathcal{D}_T$, it is enough to show

$$\mathcal{D}_{T''} \cap \{\kappa : L_{T,\beta_1}(\kappa) \geq 0\} \subset \mathcal{D}_T.$$

Note that we have $\tilde{T''} - T'' = (\tilde{T} - T) - \{\beta_1\}$, and $\mathbb{B} - \tilde{T''} = ((\mathbb{B} - \tilde{T}) - \{\beta\}) \cup \{\beta_1\}$. We also have $\epsilon_{T''}(\tau) = -\epsilon_T(\tau)$, fot $\tau \in \{\beta_1, \sigma\beta_1, ..., \sigma^{{\mu_{\beta_1}}^{-1}}\beta_1\}$, and $\epsilon_{T''}(\tau) = \epsilon_T(\tau)$, otherwise. Let $\iota : \mathbb{B} - \tilde{T} \to \mathbb{B} - \tilde{T''}$ be the bijection which is identity except at β, in which case $\iota(\beta) = \beta_1$.

Take $\kappa_0 \in \mathcal{D}_{T''} \cap \{\kappa : L_{T,\beta_1}(\kappa) \geq 0\}$. It follows that $L_{T,\tau}(\kappa_0) = L_{T'',\tau}(\kappa_0) \geq 0$, for all $\tau \in \tilde{T''} - T'' = (\tilde{T} - T) - \{\beta_1\}$, $L_{T,\beta_1}(\kappa_0) \geq 0$, and that, for $\tau \in \mathbb{B} - \tilde{T}$, either $L_{T,\tau}(\kappa_0) = L_{T'',\iota(\tau)}(\kappa_0)$ (when $\tau \notin \mathbb{B}_\mathfrak{p}$), or there is $N_\tau \geq 0$ such that $L_{T,\tau}(\kappa_0) = L_{T'',\iota(\tau)}(\kappa_0) + 2p^{N_\tau} L_{T,\beta_1}(\kappa_0) \geq 0$. Proposition 6.1 implies that $\kappa_0 \in \mathcal{D}_T$.

Case 2: $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} = \emptyset$. Recall that $\tilde{T}_\mathfrak{p} - T_\mathfrak{p} = \{\beta_1, \cdots, \beta_r\}$. We have

$$\mathcal{C}_T \subset \{\kappa \in \mathbb{Q}^\mathbb{B} : L_{T,\beta_i}(\kappa) \geq 0, 1 \leq i \leq r\},$$

by Corollary 7.5. Let $T' = T \cup \{\beta_1\}$. Therefore, Lemma 7.1 applied with $\mathbf{h}_{\beta_1}(T)$ implies that

$$\mathcal{C}_T \subseteq \{\kappa \in \mathcal{C}_{T'} + \langle h_{\beta_1} \rangle : L_{T,\beta_i}(\kappa) \geq 0, 1 \leq i \leq r\}.$$

By induction, it is enough to show that $\{\kappa \in \mathcal{D}_{T'} + \langle h_{\beta_1} \rangle : L_{T,\beta_i}(\kappa) \geq 0, 1 \leq i \leq r\} \subset \mathcal{D}_T$. Since T, T' only differ at $\mathfrak{p}$, this is equivalent to $\{\kappa \in \mathcal{D}_{T'_\mathfrak{p}} + \langle h_{\beta_1} \rangle : L_{T,\beta_i}(\kappa) \geq 0, 1 \leq i \leq r\} \subset \mathcal{D}_{T_\mathfrak{p}}$, which is clear.

□

We now summarize these results and introduce some new notation. Let $T \subseteq \mathbb{B}$. First assume $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} \neq \emptyset$. For all $\beta \in \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}$, we set $f_\beta(T) = h^\beta_{\sigma^{n_\beta}\beta}$, and $\mathbf{f}_\beta(T) = \mathbf{h}^\beta_{\sigma^{n_\beta}\beta}(T)$. For $\beta \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}$, we set $f_\beta(T) = -b^\beta_{\sigma^{n_\beta}\beta}$, and $\mathbf{f}_\beta(T) = \mathbf{h}^{\sigma\beta}_{\sigma^{n_\beta}\beta}(T)\mathbf{b}_{\sigma\beta}(T)^{-p^{n_\beta}-1}$. If $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} = \emptyset$, for $\beta \in \mathbb{B}_\mathfrak{p} - T_\mathfrak{p}$, we set $f_\beta(T) = -(1 + p^{|\mathbb{B}_\mathfrak{p}|})e_\beta$, and $\mathbf{f}_\beta(T) = \mathbf{h}^{\sigma\beta}_\beta(T)(\mathbf{b}_{\sigma\beta}(T))^{-p^{|\mathbb{B}_\mathfrak{p}|}-1}$. By Theorem 7.6, for any $T \subseteq \mathbb{B}$, we have

$$\mathcal{C}_T = \langle f_\beta(T) : \beta \in \mathbb{B} - T \rangle + \langle \pm b_\beta : \beta \in T \rangle,$$

and $0 \neq \mathbf{f}_\beta(T) \in H^0(X_T, \omega^{f_\beta(T)})$, for all $\beta \in \mathbb{B} - T$.

Lemma 4.2 implies that for each $\beta \in \mathbb{B} - T$, the zero locus of $\mathbf{f}_\beta(T)$ intersects every connected component of X_T in a smooth divisor. It follows that if C is a connected component of X_T, then

$$\mathcal{C}(C, \omega^\bullet) = \mathcal{C}(X_T, \omega^\bullet) = \mathcal{C}_T.$$

Remark 7.7. We defined the cone of weights on X_T in a way that amounts to the sum of the cones of weights of its connected components. An alternative definition would have been to consider the intersection of the cones of weights of the connected components of X_T, in which case, Lemma 7.1 would not generally hold true. However, the two notions coincide for X_T by the discussion above.

8. Minimal cones

In this section, we determine the minimal cones of strata in the spirit of [2, 3], and deduce automatic divisibility results. We will first define a stratification on X_T, and study its properties which we need in the determination of the minimal cone of X_T.

8.1. **A stratification on X_T.** We begin by some preparation. In the rest of this paper, we frequently use the notation introduced at the end of §7.

Proposition 8.1. *Let $T \subseteq \mathbb{B}$, $\mathfrak{p}|p$, and $\beta \in \mathbb{B}_{\mathfrak{p}} - T_{\mathfrak{p}}$.*

(1) *If $\tilde{T}_{\mathfrak{p}} \neq \mathbb{B}_{\mathfrak{p}}$, then, on $X_T \cap Z(\mathbf{f}_\beta(T))$, we have $\omega_{\sigma^{n_\beta}\beta} \cong {\omega_\beta}^{(-p)^{n_\beta}}$, and on $X_T - Z(\mathbf{f}_\beta(T))$, we have $\omega_{\sigma^{n_\beta}\beta} \cong {\omega_\beta}^{-(-p)^{n_\beta}}$.*

(2) *If $\tilde{T}_{\mathfrak{p}} = \mathbb{B}_{\mathfrak{p}}$, then, on $X_T \cap Z(\mathbf{f}_\beta(T))$, we have $\omega_\beta^{(-p)^{|\mathbb{B}_{\mathfrak{p}}|}-1} \cong \mathcal{O}_{X_T \cap Z(\mathbf{f}_\beta(T))}$. We also have $\omega_\beta^{-(-p)^{|\mathbb{B}_{\mathfrak{p}}|}-1} \cong \mathcal{O}_{X_T - Z(\mathbf{f}_\beta(T))}$ on $X_T - Z(\mathbf{f}_\beta(T))$.*

Proof. In case (1), we write $\{\sigma^i\beta : 0 < i < n_\beta\} \cap (\tilde{T} - T) = \{\beta_1, ..., \beta_t\}$. Let $T^0 = T \cup \{\beta_1, ..., \beta_t\}$. The map $\pi_{T,T^0} : X_T \to X_{T^0}$ restricts to give a morphism

$$\pi_{T,T^0} : X_T \cap Z(\mathbf{f}_\beta(T)) \to X_{T^0} \cap Z(\mathbf{f}_\beta(T^0)),$$

since, by the second statement in Proposition 4.1, we have $Z(\pi_{T,T^0}^* \mathbf{f}_\beta(T^0)) = Z(\mathbf{f}_\beta(T))$. In particular, $\pi_{T,T^0}^* \omega_\tau = \omega_\tau$ on $X_T \cap Z(\mathbf{f}_\beta(T))$, for $\tau = \beta, \sigma^{n_\beta}\beta$, by the properties of π_{T,T^0} discussed in §3 . The first claim now follows since $\prod_{i=1}^{n_\beta} \mathbf{b}_{\sigma^i\beta}(T^0 \cup \{\sigma^{n_\beta}\beta\})^{-(-p)^{n_\beta - i}}$ is a nowhere vanishing section of $\omega_{\sigma^{n_\beta}\beta}^{-1} \otimes {\omega_\beta}^{(-p)^{n_\beta}}$ on

$$X_{T^0} \cap Z(\mathbf{f}_\beta(T^0)) = X_{T^0} \cap Z(\mathbf{h}_{\sigma^{n_\beta}\beta}(T^0)) = X_{T^0 \cup \{\sigma^{n_\beta}\beta\}}.$$

Similarly, we have an induced morphism

$$\pi_{T,T^0} : X_T - Z(\mathbf{f}_\beta(T)) \to X_{T^0} - Z(\mathbf{f}_\beta(T^0)),$$

satisfying $\pi_{T,T^0}^* \omega_\tau = \omega_\tau$ on $X_T - Z(\mathbf{f}_\beta(T))$, for $\tau = \beta, \sigma^{n_\beta}\beta$, and the second claim follows because $\mathbf{h}_{\sigma^{n_\beta}\beta}(T^0) \prod_{i=1}^{n_\beta - 1} \mathbf{b}_{\sigma^i\beta}(T^0)^{(-p)^{n_\beta - i}}$ is a nowhere vanishing section of $\omega_{\sigma^{n_\beta}\beta}^{-1} \otimes {\omega_\beta}^{-(-p)^{n_\beta}}$ on $X_{T^0} - Z(\mathbf{f}_\beta(T^0)) = X_{T^0} - Z(\mathbf{h}_{\sigma^{n_\beta}\beta}(T^0))$.

In case (2), letting $T^0 = T \cup (\mathbb{B}_{\mathfrak{p}} - \{\beta\})$, and arguing similarly, it is enough to prove the statements over $X_{T^0} \cap Z(\mathbf{f}_\beta(T^0))$. Since

$$X_{T^0} \cap Z(\mathbf{f}_\beta(T^0)) = X_{T^0} \cap Z(\mathbf{h}_\beta(T^0)) = X_{T^0 \cup \{\beta\}},$$

and $\mathbb{B}_{\mathfrak{p}} \subset T^0 \cup \{\beta\}$, by Corollary 2.3, we have $\omega_\beta^{(-p)^{|\mathbb{B}_{\mathfrak{p}}|}-1} \cong \mathcal{O}_{X_{T^0} \cap Z(\mathbf{f}_\beta(T^0))}$. The second claim follows since $\mathbf{h}_\beta(T^0) \prod_{i=1}^{n_\beta - 1} \mathbf{b}_{\sigma^i\beta}(T^0)^{(-p)^{n_\beta - i}}$ is a nowhere vanishing section of $\omega_\beta^{-(-p)^{|\mathbb{B}_{\mathfrak{p}}|}-1}$ on $X_{T^0} - Z(\mathbf{f}_\beta(T^0)) = X_{T^0} - Z(\mathbf{h}_\beta(T^0))$.

□

Definition 8.2. Let $T \subseteq \mathbb{B}$, and $J \subseteq \mathbb{B} - T$. We define

$$X_{T,J} = X_T \cap \big(\bigcap_{\beta \in J} Z(\mathbf{f}_\beta(T)) \big),$$

$$W_{T,J} = X_{T,J} - \bigcup\nolimits_{\beta \notin J} X_{T,\{\beta\}}.$$

Proposition 8.3. *If $T \subseteq \mathbb{B}$ and $J \subseteq \mathbb{B} - T$, then $W_{T,J}$ is quasi-affine.*

Proof. Using all the isomorphisms given in Proposition 8.1 and Definition 2.2, it is easy to see that for all $\beta \in \mathbb{B}$, ω_β is a torsion line bundle over $W_{T,J}$. Since $\bigotimes_{\beta \in \mathbb{B}} \omega_\beta$ is ample on X, it follows that its restriction to $W_{T,J}$ is both ample and torsion, which implies that $W_{T,J}$ is quasi-affine (since it is quasi-compact). □

Corollary 8.4. *Let $T \subset \mathbb{B}$, $J \subset \mathbb{B} - T$, and assume $X_{T,J}$ is nonempty. Then, for any $\beta \notin T \cup J$, $X_{T,\{\beta\}}$ intersects every connected component of $X_{T,J}$.*

Proof. The case $T = \emptyset$ has been proved in Corollary 5.7 of [3]. We assume $T \neq \emptyset$, and, therefore, X_T is proper. Let $J \subset \mathbb{B} - T$ be a subset of maximal cardinality for which the claim fails to be true (in particular $J \neq \mathbb{B} - T$). Let C be a connected component of $X_{T,J}$ such that $C \cap X_{T,\{\beta\}} = \emptyset$ for some $\beta \notin T \cup J$. The maximality assumption implies that $C \cap X_{T,\{\tau\}} = \emptyset$, for all $\tau \notin T \cup J$. It follows that $C \subset W_{T,J}$. Since $W_{T,J}$ is quasi-compact, Proposition 8.3 implies that C is quasi-affine. Since C is also proper, it follows that C is finite. This is a contradiction, since $X_{T,J}$ is cut out from X_T by the vanishing of $|J|$ sections, and, hence, every connected component of X_T has codimension at most $|J| < |\mathbb{B} - T| = \dim X_T$.

□

Proposition 8.5. *Let $T \subseteq \mathbb{B}$, $J \subseteq \mathbb{B} - T$. We have*

(1) $X_{T,J} = \overline{W}_{T,J} = \bigcup_{J' \supseteq J} W_{T,J'}$.
(2) *Each $W_{T,J}$ (and hence $X_{T,J}$) is non-empty and equidimensional of dimension $|\mathbb{B} - (T \cup J)|$.*

Proof. This can be proved exactly as in Proposition 5.8 of [3] using proposition 8.3, and Corollary 8.4. □

Remark 8.6. The above construction gives a stratification on (the connected components of) X_T which is different from the one induced by the Goren-Oort stratification on X in general.

8.2. **The minimal cone of X_T.** In this section, we define and determine the minimal cone of X_T. We begin by stating some consequences of the results in the previous section that we will need in our arguments.

Lemma 8.7. *Let $T \subseteq \mathbb{B}$, and $J \subseteq \mathbb{B} - T$. Let $\beta \in \mathbb{B} - (T \cup J)$, and $T' = T \cup \{\beta\}$. Then $X_{T',J}$ intersects every connected component of $X_{T,J}$.*

Proof. Let C be a connected component of $X_{T,J}$. By Corollary 8.4 and Proposition 8.5, we know that $Z := C \cap X_{\mathbb{B}-(J\cup\{\beta\}),J}$ is nonempty, projective, and of pure dimension 1. Assume $X_{T',J} \cap C = \emptyset$. Then $Z \subseteq X_{T,J} - X_{T,J'}$. In particular, $Z \cap X_{T,J'} \neq \emptyset$, which implies that $\omega_\beta \cong \omega_{\sigma^{-1}\beta}^{-p}$ on Z. This, together with all the isomorphisms obtained via Proposition 8.1, imply that ω_τ is torsion on Z, for all $\tau \in \mathbb{B}$. In particular, we find that $\bigotimes_{\tau\in\mathbb{B}} \omega_\tau$ is both ample and torsion on Z. This implies that Z is quasi-affine, which is a contradiction. □

Lemma 8.8. *Let $T \subseteq \mathbb{B}$, $\beta \in \mathbb{B} - T$, and C a connected component of $X_T \cap Z(\mathbf{f}_\beta(T))$. We have*

$$\langle f_\tau(T) : \tau \in \mathbb{B} - (T \cup \{\beta\})\rangle + \langle \pm b_\tau : \tau \in T\rangle \subset \mathcal{C}(C, \omega^\bullet).$$

In particular, $\langle f_\tau(T) : \tau \in \mathbb{B} - (T \cup \{\beta\})\rangle + \langle \pm b_\tau : \tau \in T\rangle \subset \mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet)$.

Proof. Clearly, for all $\tau \in T$, we have $\pm b_\tau \in \mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet)$. If $\tau \in \mathbb{B} - (T \cup \{\beta\})$, then $\mathbf{f}_\tau(T)$ restricts to a nonzero section of $\omega^{f_\tau(T)}$ on any connected component of $X_T \cap Z(\mathbf{f}_\beta(T)) = X_{T,\{\beta\}}$, by Lemma 8.7. □

Given $T \subset \mathbb{B}$, and $\mathfrak{p}|p$, we define $\mathbb{B}_\mathfrak{p}(T)$ as follows:

$$\mathbb{B}_\mathfrak{p}(T) = \begin{cases} \mathbb{B}_\mathfrak{p} - T_\mathfrak{p} & \text{if } |\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p}| > 1, \\ \mathbb{B}_\mathfrak{p} - (T_\mathfrak{p} \cup \{\beta\}) & \text{if } \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} = \{\beta\}, \\ \emptyset & \text{if } \mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} = \emptyset. \end{cases}$$

We define $\mathbb{B}(T) = \bigcup_{\mathfrak{p}|p} \mathbb{B}_{\mathfrak{p}}(T)$. Note that $\beta \in \mathbb{B}(T)$ is equivalent to the condition that n_β is defined, and $\sigma^{n_\beta}\beta \neq \beta$.

Definition 8.9. For any $\beta \in \mathbb{B}(T)$, define $\epsilon_{T,f_\beta} : \mathbb{B} \to \{-1, 0, 1\}$ as

$$\epsilon_{T,f_\beta}(\tau) = \begin{cases} -\epsilon_T(\tau) & \text{if } \tau \in \{\beta, \sigma\beta, ..., \sigma^{\mu_\beta - 1}\beta\}, \\ \epsilon_T(\tau) & \text{otherwise.} \end{cases}$$

Given $\beta \in \mathbb{B}(T)$, using notation from §6, for $\tau \in \mathbb{B} - (T \cup \{\sigma^{n_\beta}\beta\})$, we define

$$L_{T,f_\beta,\tau} = \begin{cases} L^{\epsilon_{T,f_\beta}}[\sigma^{\mu_\tau}\tau] & \text{if } \beta \neq \tau \in \mathbb{B} - \tilde{T}, \\ L_{T,\tau} & \text{if } \beta \neq \tau \in \tilde{T} - T, \\ L^{\epsilon_{T,f_\beta}}[\beta, \sigma^{-1}\tilde{\beta}] & \text{if } \beta = \tau \in \mathbb{B} - \tilde{T}, \\ L^{\epsilon_{T,f_\beta}}[\tilde{\beta}] & \text{if } \beta = \tau \in \tilde{T} - T, \end{cases}$$

where, $\beta' = \sigma^{n_\beta}\beta$, and $\tilde{\beta} = \sigma^{\mu_{\beta'}}\beta'$. Note that if $\beta \in \mathbb{B}_{\mathfrak{p}}$, then, $L_{T,f_\beta,\tau} = L_{T,\tau}$, for all $\tau \notin \mathbb{B}_{\mathfrak{p}}$. Finally, for $\beta \in \mathbb{B}(T)$, we define

$$\mathcal{D}_{T,f_\beta} = \{\kappa \in \mathbb{Q}^{\mathbb{B}} : L_{T,f_\beta,\tau}(\kappa) \geq 0 : \forall \tau \in \mathbb{B} - (T \cup \{\sigma^{n_\beta}\beta\})\}.$$

Theorem 8.10. *Let $T \subset \mathbb{B}$, and $\beta \in \mathbb{B} - T$.*

1. *If $\beta \in \mathbb{B}(T)$, we have*
   (a) $\dim H^0(C, \omega^{f_\beta(T)}) = 1$, *for any connected component C of X_T.*
   (b) $\mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet) = \mathcal{D}_{T,f_\beta}$.
2. *If $\beta \notin \mathbb{B}(T)$, we have $\mathcal{C}_T \subseteq \mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet)$.*

Proof. First, we show that for $\beta \in \mathbb{B}(T)$, (1)(b) implies (1)(a). Let C be a connected component of X_T, and $\mathbf{f} \in H^0(C, \omega^{f_\beta(T)})$. Let $\tilde{\mathbf{f}}$ denote the extension of $\mathbf{f}$ to X_T by setting it zero on the rest of the connected components. Since $\beta \in \mathbb{B}(T)$, we can easily see that there is $\delta \geq 0$ such that $L_{T,f_\beta,\beta}(f_\beta(T)) = -2p^{n_\beta+\delta} < 0$, which implies that $f_\beta(T) \notin \mathcal{D}_{T,f_\beta}$. By (1)(b), it follows that $f_\beta(T) \notin \mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)))$, and so the restriction of $\tilde{\mathbf{f}}$ to $X_T \cap Z(\mathbf{f}_\beta(T))$ is zero. This means $\mathbf{f}$ is divisible by the restriction of $\mathbf{f}_\beta(T)$ to C (which is nonzero by the discussion at the end of §7), from which the claim follows.

Next, we prove (1)(b). We begin by show that $\mathcal{D}_{T,f_\beta} = \mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet)$, for all $\beta \in \mathbb{B}(T)$, if we have $\{\sigma^i\beta : 0 < i < n_\beta\} \cap (\tilde{T} - T) = \emptyset$. In this case, Proposition 4.1 tells us that $\mathbf{f}_\beta(T) = \mathbf{h}_{\sigma^{n_\beta}\beta}(T) \prod_{0<i<n_\beta} \mathbf{b}_{\sigma^i\beta}(T)^{(-p)^i}$, since, by assumption, $\{\sigma^i\beta : 0 \leq i < n_\beta\} \subset T$, and $T^0 = T$ (in the notation of the proposition). It follows that $X_T \cap Z(\mathbf{f}_\beta(T)) = X_T \cap Z(\mathbf{h}_{\sigma^{n_\beta}\beta}(T)) = X_{T \cup \{\sigma^{n_\beta}\beta\}}$, from which the claim follows by Proposition 6.1 and Theorem 7.6, noting that in this case, $L_{T,f_\beta,\tau} = L_{T\cup\{\sigma^{n_\beta}\beta\},\tau}$, for all $\tau \in \mathbb{B} - (T \cup \{\sigma^{n_\beta}\beta\})$.

We now consider a general T, and prove $\mathcal{D}_{T,f_\beta} \subset \mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet)$, for all $\beta \in \mathbb{B}(T)$, using an induction on the cardinality of $\{\sigma^i\beta : 0 < i < n_\beta\} \cap (\tilde{T} - T)$. The base case was proved above. Now assume $\{\sigma^i\beta : 0 < i < n_\beta\} \cap (\tilde{T} - T) = \{\sigma^{r_i}\beta : 1 \leq i \leq t\}$, where $0 < r_1 < ... < r_t < n_\beta$, and $t \geq 1$. Let $\kappa \in \mathcal{D}_{T,f_\beta}$. By discussions in §5 (See Equation 5.1), for $r_t < i < n_\beta$, we can find $\lambda_{\sigma^i\beta} \in \mathbb{Q}$ such that

$$\kappa' := \kappa + \sum_{r_t<i<n_\beta} \lambda_{\sigma^i\beta} b_{\sigma^i\beta} - p^{r_t - n_\beta} L_{T,\sigma^{r_t}\beta}(\kappa) f_{\sigma^{r_t}\beta}(T),$$

satisfies $k'_\tau = 0$, if $\tau = \sigma^i\beta$ for $r_t \le i < n_\beta$, and $k'_{\sigma^{n_\beta}\beta} = k_{\sigma^{n_\beta}\beta} + p^{r_t - n_\beta} L_{T,\sigma^{r_t}\beta}(\kappa)$.

Since $\kappa \in \mathcal{D}_{T,f_\beta}$, we have $L_{T,\sigma^{r_t}\beta}(\kappa) = L_{T,f_\beta,\sigma^{r_t}\beta}(\kappa) \ge 0$. Lemma 8.8 now implies that $\kappa \in \mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet)$ follows from $\kappa' \in \mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet)$. We let $T' = T \cup \{\sigma^{r_t}\beta\}$, and note that $f_\beta(T') = f_\beta(T)$, meaning we have $0 \ne \mathbf{f}_\beta(T') \in H^0(X_{T'}, \omega^{f_\beta(T)})$. Furthermore, by the second statement in Proposition 4.1, we have $X_T \cap Z(\pi^*_{T,T'}\mathbf{f}_\beta(T')) = X_T \cap Z(\mathbf{f}_\beta(T))$. Therefore, $\pi_{T,T'} : X_T \to X_{T'}$ (defined in §3) restricts to give a $\mathbb{P}^1$-bundle morphism

$$\pi_{T,T'} : X_T \cap Z(\mathbf{f}_\beta(T)) \to X_{T'} \cap Z(\mathbf{f}_\beta(T')).$$

Hence, Lemma 3.2 implies that

$$\pi^*_{T,T'} : H^0(X_{T'} \cap Z(\mathbf{f}_\beta(T')), \omega^{\kappa'}) \to H^0(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^{\kappa'})$$

is an isomorphism (noting that $\mu_{\sigma^{r_t}\beta} = n_\beta - r_t$). Consequently, it is enough to prove that $\kappa' \in \mathcal{D}_{T',f_\beta} \subseteq \mathcal{C}(X_{T'} \cap Z(\mathbf{f}_\beta(T')), \omega^\bullet)$, where the inclusion holds by the induction hypothesis. This follows by checking that for all $\tau \in \mathbb{B} - (T \cup \{\sigma^{r_t}\beta, \sigma^{n_\beta}\beta\})$, we have $L_{T',f_\beta,\tau}(\kappa') = L_{T,f_\beta,\tau}(\kappa)$ (by considering the cases $t = 1$ and $t > 1$ separately).

To prove the reverse inclusion, i.e., $\mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet) \subset \mathcal{D}_{T,f_\beta}$, for $\beta \in \mathbb{B}(T)$, we again use an induction on the cardinality of $\{\sigma^i\beta : 0 < i < n_\beta\} \cap (\tilde{T} - T) = \{\sigma^{r_i}\beta : 1 \le i \le t\}$ (the base case having already been proved above). Let $T' = T \cup \{\sigma^{r_t}\beta\}$, and $T'' = T \cup \{\sigma^{r_1}\beta, ..., \sigma^{r_t}\beta\}$. Note that $f_\beta(T'') = f_\beta(T)$, and $f_{\sigma^{n_\beta}\beta}(T'') = f_{\sigma^{n_\beta}\beta}(T)$. Considering $\pi_{T,T''} : X_T \to X_{T''}$, and using an argument as in the case of $\pi_{T,T'}$ above, we see that we have a $(\mathbb{P}^1)^t$-bundle map

$$\pi_{T,T''} : Y := X_T \cap Z(\mathbf{f}_\beta(T)) \to Y'' := X_{T''} \cap Z(\mathbf{f}_\beta(T'')).$$

We want to apply Lemma 7.1 on $X_T \cap Z(\mathbf{f}_\beta(T))$ with $\mathbf{h}_{\sigma^{r_t}\beta}(T)$ and $\pi^*_{T,T''}\mathbf{f}_{\sigma^{n_\beta}\beta}(T'') = \mathbf{f}_{\sigma^{n_\beta}\beta}(T)$. These sections satisfy the conditions required in Lemma 7.1, by Lemma 8.7 (and noting that $\sigma^{n_\beta}\beta \ne \beta$, $\sigma^{r_t}\beta \ne \beta$, $\sigma^{n_\beta}\beta \notin T$). By Lemma 7.1, we have

$$\mathcal{C}(Y, \omega^\bullet) \subseteq \big(\mathcal{C}(Y \cap Z(\mathbf{h}_{\sigma^{r_t}\beta}(T)), \omega^\bullet) \cap \mathcal{C}(Y \cap Z(\pi^*_{T,T''}\mathbf{f}_{\sigma^{n_\beta}\beta}(T'')), \omega^\bullet)\big) + \langle h_{\sigma^{r_t}\beta}, f_{\sigma^{n_\beta}\beta}(T)\rangle.$$

Note that $Y \cap Z(\mathbf{h}_{\sigma^{r_t}\beta}(T)) = \big(X_T \cap Z(\mathbf{f}_\beta(T)\big) \cap Z(\mathbf{h}_{\sigma^{r_t}\beta}(T))$ can be thought of as the vanishing locus of $\mathbf{f}$, the restriction of $\mathbf{f}_\beta(T)$ to $X_T \cap Z(\mathbf{h}_{\sigma^{r_t}\beta}(T)) = X_{T'}$. The section $\mathbf{f}$ is a section of $\omega^{f_\beta(T)} = \omega^{f_\beta(T')}$. We prove it is nonzero on every connected component of $X_{T'}$. Since, by discussions in §3, X_T is a $\mathbb{P}^1$-bundle over $X_{T'}$, the connected components of X_T and $X_{T'}$ are in bijection. By Lemma 4.1 of [2], the connected components of $X_{T'}$ map onto those of X_T under the natural embedding $X'_T \to X_T$. Hence, given a connected component C' of $X_{T'}$, there is a connected component C of X_T such that $C' = C \cap X_{T'} = C \cap Z(\mathbf{h}_{\sigma^{r_t}\beta}(T))$. If $\mathbf{f}$ is zero on C', it follows that $\mathbf{f}_\beta(T)$ is divisible by $\mathbf{h}_{\sigma^{r_t}\beta}(T)$ over C. It follows that $\lambda := f_\beta(T) - h_{\sigma^{r_t}\beta} \in \mathcal{C}(C, \omega^\bullet)$, being the weight of the nonzero section $\mathbf{f}_\beta(T)/\mathbf{h}_{\sigma^{r_t}\beta}(T)$ on C. Since $L_{T,\sigma^{r_t}\beta}(\lambda) = -1 < 0$, Proposition 6.1 implies that $\lambda \notin \mathcal{D}_T = \mathcal{C}_T = \mathcal{C}(C, \omega^\bullet)$. (For the last equality, see discussions at the end of §7). This is a contradiction.

Therefore, $\mathbf{f}$ is nonzero on every connected comopnent of $X_{T'}$, and by (1)(a) for T' (which follows from (1)(b) for T', by the induction hypothesis) $\mathbf{f}$ must be a nonzero scalar multiple of $\mathbf{f}_\beta(T')$ on each connected component of X'_T. We deduce that

$$Y \cap Z(\mathbf{h}_{\sigma^{r_t}\beta}(T)) = X_{T'} \cap Z(\mathbf{f}_\beta(T')).$$

By Proposition 7.3, we have

$$\mathcal{C}(Y \cap Z(\pi^*_{T,T'}\mathbf{f}_{\sigma^{n_\beta}\beta}(T'')), \omega^\bullet) \subseteq \{\kappa : L_{T,\sigma^{r_i}\beta}(\kappa) \ge 0,\ \forall\ 1 \le i \le t\},$$

and, by the induction hypothesis, we have

$$\mathcal{C}(Y \cap Z(\mathbf{h}_{\sigma^{r_t}\beta}(T)), \omega^\bullet) = \mathcal{C}(X_{T'} \cap Z(\mathbf{f}_\beta(T')), \omega^\bullet) \subseteq \mathcal{D}_{T',f_\beta}.$$

Putting all these together, we have

$$\mathcal{C}(Y, \omega^\bullet) \subseteq \big(\mathcal{D}_{T',f_\beta} \cap \{\kappa : L_{T,\sigma^{r_i}\beta}(\kappa) \geq 0,\ \forall\ 1 \leq i \leq t\}\big) + \langle h_{\sigma^{r_t}\beta}, f_{\sigma^{n_\beta}\beta}(T)\rangle.$$

It can be checked that $h_{\sigma^{r_t}\beta}, f_{\sigma^{n_\beta}\beta}(T) \in \mathcal{D}_{T,f_\beta}$. Therefore, it is enough to show

$$\mathcal{D}_{T',f_\beta} \cap \{\kappa : L_{T,\sigma^{r_i}\beta}(\kappa) \geq 0,\ \forall\ 1 \leq i \leq t\}\big) \subset \mathcal{D}_{T,f_\beta}.$$

First assume $t > 1$, in which case $\epsilon_{T,f_\beta} = \epsilon_{T',f_\beta}$, and from which it is easy to check that $L_{T,f_\beta,\tau}(\kappa) = L_{T',f_\beta,\tau}(\kappa) \geq 0$ for all $\tau \in \mathbb{B} - (T \cup \{\sigma^{n_\beta}\beta\})$ such that $\tau \notin \{\sigma^{r_t}\beta, \sigma^{r_{t-1}}\beta\}$. The result follows since the two remaining conditions for $\kappa \in \mathcal{D}_{T,f_\beta}$ are also satisfied: $L_{T,f_\beta,\tau}(\kappa) = L_{T,\tau}(\kappa) \geq 0$ for $\tau \in \{\sigma^{r_t}\beta, \sigma^{r_{t-1}}\beta\}$. Now assume $t = 1$. In this case, $\epsilon_{T,f_\beta}(\tau) = -\epsilon_{T',f_\beta}(\tau)$ if $\tau \in \{\sigma^{r_1}\beta, ..., \sigma^{n_\beta - 1}\beta\}$, and $\epsilon_{T,f_\beta}(\tau) = \epsilon_{T',f_\beta}(\tau)$ otherwise. An examination of the defintion shows that for all $\sigma^{r_1}\beta \neq \tau \in \mathbb{B} - (T \cup \{\sigma^{n_\beta}\beta\})$, either $L_{T,f_\beta,\tau} = L_{T',f_\beta,\tau}$, or there is $N_\tau \geq 0$, such that $L_{T,f_\beta,\tau} - L_{T',f_\beta,\tau} = 2p^{N_\tau} L_{\sigma^{r_1}\beta}(\kappa) \geq 0$. The last condition for $\kappa \in \mathcal{D}_{T,f_\beta}$ is $L_{T,f_\beta,\sigma^{r_1}\beta}(\kappa) = L_{T,\sigma^{r_1}\beta}(\kappa) \geq 0$, which also holds by the assumption.

We now prove (2). By assumption, either $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} = \emptyset$ (and $\beta \in \tilde{T}_\mathfrak{p} - T_\mathfrak{p}$), or $\mathbb{B}_\mathfrak{p} - \tilde{T}_\mathfrak{p} = \{\beta\}$. By Lemma 8.8, it is enough to show that $f_\beta(T) \in \mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet)$. Let $T' = T \cup (\mathbb{B}_\mathfrak{p} - \{\beta\})$, so that, by Proposition 4.1, we have

$$\pi_{T,T'} : X_T \cap Z(\mathbf{f}_\beta(T)) \to X_{T'} \cap Z(\mathbf{f}_\beta(T'))$$

is the restriction of $\pi_{T,T'} : X_T \to X_{T'}$ to $X_{T'} \cap Z(\mathbf{f}_\beta(T'))$. By Lemma 3.2, it is enough to show that $f_\beta(T) \in \mathcal{C}(X_{T'} \cap Z(\mathbf{f}_\beta(T')), \omega^\bullet)$. But, by Proposition 4.1, we have $X_{T'} \cap Z(\mathbf{f}_\beta(T')) = X_{T'} \cap Z(\mathbf{h}_\beta(T')) = X_{T' \cup \{\beta\}}$, and $\Sigma_{\tau \in \mathbb{B}_\mathfrak{p}} \mathbb{Q} e_\tau \subseteq \mathcal{C}_{T' \cup \{\beta\}}$, as $\mathbb{B}_\mathfrak{p} \subseteq T' \cup \{\beta\}$. □

Corollary 8.11. *Let $T \subseteq \mathbb{B}$. Assume $\mathbf{f} \in H^0(X_T, \omega^\kappa)$, and $L_{T,f_\beta,\tau}(\kappa) < 0$, for some $\beta \in \mathbb{B}(T)$, $\tau \in \mathbb{B} - (T \cup \{\sigma^{n_\beta}\beta\})$. Then $\mathbf{f}$ is divisible by $\mathbf{f}_\beta(T)$.*

Proof. By Theorem 8.10, the assumption implies that $\kappa \notin \mathcal{C}(X_T \cap Z(\mathbf{f}_\beta(T)), \omega^\bullet)$. Hence, the restriction of $\mathbf{f}$ to $X_T \cap Z(\mathbf{f}_\beta(T))$ is zero, which implies that $\mathbf{f}$ is divisible by $\mathbf{f}_\beta(T)$. □

Definition 8.12. Let $T \subseteq \mathbb{B}$. The minimal cone of X_T is defined to be

$$\mathcal{C}_T^{\min} = i_T(\mathcal{C}_T \cap (\bigcap_{\beta \in \mathbb{B}(T)} \mathcal{D}_{T,f_\beta})) \subseteq \mathcal{C}_T^{T-\mathrm{red}} \subseteq \mathbb{Q}^{\mathbb{B}-T},$$

where $i_T : \mathbb{Q}^\mathbb{B} \to \mathbb{Q}^{\mathbb{B}-T}$ is defined in §5). Note that for all $\beta \in T$, we have

$$\mathrm{Ker}(i_T) = \langle \pm b_\beta : \beta \in T\rangle \subseteq \mathcal{C}_T \cap (\bigcap_{\beta \in \mathbb{B}(T)} \mathcal{D}_{T,f_\beta}).$$

It follows that $\mathcal{C}_T \cap (\bigcap_{\beta \in \mathbb{B}(T)} \mathcal{D}_{T,f_\beta}) = i_T^{-1}(\mathcal{C}_T^{\min}) = j_T(C_T^{\min}) + \langle \pm b_\beta : \beta \in T\rangle$.

Proposition 8.13. *Let $T \subseteq \mathbb{B}$ be such that $\tilde{T} = T$. Then*

$$\mathcal{C}_T^{\min} = \{\sum_\beta \ell_\beta e_\beta \in \mathbb{Q}^{\mathbb{B}-T} : p^{n_\beta} \ell_{\sigma^{n_\beta}\beta} \geq \ell_\beta, \forall \beta \in \mathbb{B} - T\}.$$

Proof. The projection $i_T : \mathbb{Q}^{\mathbb{B}} \to \mathbb{Q}^{\mathbb{B}-\mathbb{T}}$ is given by $i_T(\sum_{\beta\in\mathbb{B}} k_\beta e_\beta) = \sum_{\beta\in\mathbb{B}-T} \ell_\beta e_\beta$, where, for $\beta \in \mathbb{B} - T$, we have $\ell_\beta = \sum_{i=0}^{\mu_\beta - 1}(-p)^i k_{\sigma^i\beta}$. Note that for $\beta \in \mathbb{B} - \tilde{T} = \mathbb{B} - T$, we have

$$L_{T,f_\beta,\beta}(\sum_{\beta\in\mathbb{B}} k_\beta e_\beta) = p^{n_\beta}\ell_{\sigma^{n_\beta}\beta} - \ell_\beta.$$

This proves one inclusion. For the reverse inclusion, we need to show that if $\kappa = \sum_{\beta\in\mathbb{B}} k_\beta e_\beta$ is such that $L_{T,f_\beta,\beta}(\sum_{\beta\in\mathbb{B}} k_\beta e_\beta) = p^{n_\beta}\ell_{\sigma^{n_\beta}\beta} - \ell_\beta \geq 0$, for all $\beta \in \mathbb{B} - T$, then $\kappa \in \mathcal{C}_T \cap (\bigcap_{\beta\in\mathbb{B}-T} \mathcal{D}_{T,f_\beta})$. Note that these conditions imply $\ell_\beta \geq 0$ for all $\beta \in \mathbb{B} - T$. For $\mathfrak{p}|p$, and $\tau \in \mathbb{B}_\mathfrak{p} - T_\mathfrak{p}$, set $\bar{\sigma}\tau := \sigma^{n_\tau}\tau \in \mathbb{B}_\mathfrak{p} - T_\mathfrak{p}$, and for $1 \leq i \leq |\mathbb{B}_\mathfrak{p} - T_\mathfrak{p}|$, let $0 < n_i(\tau) \leq |\mathbb{B}_\mathfrak{p}|$ be such that $\bar{\sigma}^i\tau = \sigma^{n_i(\tau)}\tau$. Then,

$$p^{n_1(\tau)}L_{T,\tau}(\kappa) = \sum_{i=1}^{|\mathbb{B}_\mathfrak{p}-T_\mathfrak{p}|} p^{n_i(\tau)}\ell_{\bar{\sigma}^i\tau} \geq 0.$$

It remains to show that $L_{T,f_\beta,\tau}(\kappa) \geq 0$, for all $\beta \in \mathbb{B}(T) = \mathbb{B} - T$ and $\beta \neq \tau \in \mathbb{B} - (T \cup \{\sigma^{n_\beta}\beta\})$. If $\beta \in \mathbb{B}_\mathfrak{p}$ and $\tau \notin \mathbb{B}_\mathfrak{p}$, we have $L_{T,f_\beta,\tau}(\kappa) = L_{T,\tau}(\kappa) \geq 0$ from above. Hence, we can assume $\tau \in \mathbb{B}_\mathfrak{p}$, and pick $1 \leq j \leq |B_\mathfrak{p} - T_\mathfrak{p}|$ such that $\bar{\sigma}^j\tau = \beta$. Let $I = \{1, ..., |\mathbb{B}_\mathfrak{p} - T_\mathfrak{p}|\} - \{j, j+1\}$. Then

$$p^{n_1(\tau)}L_{T,f_\beta,\tau}(\kappa) = \sum_{i\in I} p^{n_i(\tau)}\ell_{\bar{\sigma}^i\tau} + p^{n_j(\tau)}(p^{n_\beta}\ell_{\sigma^{n_\beta}\beta} - \ell_\beta) \geq 0.$$

□

Remark 8.14. For a general T, set $\mathcal{C}_T^{\min,0} = \iota_T(\{\kappa \in \mathcal{C}_T : L_{T,\beta,f_\beta}(\kappa) \geq 0, \forall\beta \in \mathbb{B}(T)\})$. It is clear that $\mathcal{C}_T^{\min} \subseteq \mathcal{C}_T^{\min,0}$. Proposition 8.13 shows that $\mathcal{C}_T^{\min} = \mathcal{C}_T^{\min,0}$ if $\tilde{T} = T$. We expect this equality to be true for all $T \subseteq \mathbb{B}$.

9. Minimal weights

Let $T \subseteq \mathbb{B}$ and $0 \neq \mathbf{f} \in H^0(X_T, \omega^\kappa)$. For each $\beta \notin T$, we let a_β be the maximal power of $\mathbf{f}_\beta(T)$ that divides $\mathbf{f}$ (which exists by Lemma 7.2 of [3]). It follows from part (2) of Proposition 8.5 that for any $\beta, \beta' \notin T$, the strata $X'_T \cap Z(\mathbf{f}_\beta(T))$ and $X'_T \cap Z(\mathbf{f}_{\beta'}(T))$ have no common irreducible components. This implies that $\mathbf{f}$ is divisible by $\prod_{\beta\in\mathbb{B}(T)} \mathbf{f}_\beta(T)^{a_\beta}$. Hence, we can write

$$\mathbf{f} = \mathbf{f}_0 \prod_{\beta\notin T} \mathbf{f}_\beta(T)^{a_\beta}$$

for some $\mathbf{f}_0 \in H^0(X_T, \omega^{\kappa_0})$, where $\kappa_0 = \kappa - \sum_{\beta\notin T} a_\beta f_\beta(T)$. It follows from the definition of the a_β that $\mathbf{f_0}$ is not divisible by $\mathbf{f}_\beta(T)$ for any $\beta \notin T$, and that $i_T(\kappa_0) \in \mathcal{C}_T^{T-\mathrm{red}}$, where $i_T : \mathbb{Q}^{\mathbb{B}} \to \mathbb{Q}^{\mathbb{B}-T}$ is defined in (5.1).

Definition 9.1. For $0 \neq \mathbf{f} \in H^0(X_T, \omega^\kappa)$, we define the *minimal weight* of $\mathbf{f}$ to be

$$\Phi(\mathbf{f}) = i_T(\kappa_0) \in \mathbb{Z}^{\mathbb{B}-T}.$$

Proposition 9.2. *Let $T \subseteq \mathbb{B}$, and $0 \neq \mathbf{f} \in H^0(X_T, \omega^\kappa)$. Then we have $\Phi(\mathbf{f}) \in \mathcal{C}_T^{\min}$.*

Proof. Write $\mathbf{f} = \mathbf{f}_0 \prod_{\beta\notin T} \mathbf{f}_\beta(T)^{a_\beta}$, where a_β is as above, $\mathbf{f}_0 \in H^0(X'_T, \delta^{\xi_0}\omega^{\kappa_0})$, and $i_T(\kappa_0) = \Phi(\mathbf{f}) \in \mathcal{C}_T^{\mathrm{red}}$. If $\Phi(\mathbf{f}) \notin \mathcal{C}_T^{\min}$, then $\kappa_0 \notin i_T^{-1}(\mathcal{C}_T^{\min}) = \mathcal{C}_T \cap (\bigcap_{\beta\in\mathbb{B}(T)} \mathcal{D}_{T,f_\beta})$. Since $\kappa_0 \in \mathcal{C}_T$, it follows that $\kappa_0 \notin \mathcal{D}_{T,f_\beta}$, for some $\beta \in \mathbb{B}(T)$. Hence, there exist $\beta \in \mathbb{B}(T)$, and $\tau \in \mathbb{B} - (T \cup \{\sigma^{n_\beta}\beta\})$, such that $L_{T,f_\beta,\tau}(\kappa_0) < 0$. By Corollary 8.11, it follows that $\mathbf{f}_0$ is divisible by $\mathbf{f}_\beta(T)$. This contradicts the definition of $\mathbf{f}_0$. □

10. The $\mathrm{Res}_{F/\mathbb{Q}}\mathrm{GL}_2$ setting

In this section, we explain how to deduce results analogous to those of preceding sections, but in the context of Shimura varieties for the group $\mathrm{Res}_{F/\mathbb{Q}}\mathrm{GL}_2$. In so doing, we also account for the (prime-to-p) Hecke action.

10.1. **Hilbert modular varieties and automorphic line bundles.** First consider the scheme

$$\widetilde{X}_K := \coprod_{\epsilon} X_K^{\epsilon},$$

where the disjoint union runs over all $\epsilon \in (\mathbb{A}_F^{\infty,p})^\times / \det(K^p)(\widehat{\mathbb{Z}}^{(p)})^\times$(and K, as always, is a sufficiently small open compact subgroup of $\mathrm{GL}_2(\mathbb{A}_F^\infty)$ with $K_p = \mathrm{GL}_2(\mathcal{O}_{F,p})$). Let $\mathcal{O}_{F,(p)}$ denote the localization of $\mathcal{O}_F$ at the prime p of $\mathbb{Z}$, and $\mathcal{O}_{F,(p),+}^\times$ the subgroup of $\mathcal{O}_{F,(p)}^\times$ consisting of totally positive elements. We have a natural action of $\mathcal{O}_{F,(p),+}^\times$ on $\widetilde{X}_K$ defined by $\nu \cdot (A/S, \iota, \lambda, \alpha) = (A/S, \iota, \nu\lambda, \alpha)$ for $\nu \in \mathcal{O}_{F,(p),+}^\times$, sending $X_K^\epsilon(S)$ to $X_K^{\nu\epsilon}(S)$. Note that the action factors through $\mathcal{O}_{F,(p),+}^\times/(\mathcal{O}_F^\times \cap K)^2$. Moreover the resulting action is free for K sufficiently small (see [5, Lemma 2.4.1]), and the quotient $\widetilde{X}_K/\mathcal{O}_{F,(p),+}^\times$ is smooth of relative dimension d and finite type over $\mathbb{Z}_{(p)}$. Indeed it admits a finite étale cover by the disjoint union of the X_K^ϵ, where ϵ runs over representatives of the finite quotient

$$(\mathbb{A}_F^{\infty,p})^\times/\mathcal{O}_{F,(p),+}^\times \det(K^p)(\widehat{\mathbb{Z}}^{(p)})^\times = \mathbb{A}_F^\times/F^\times F_{\infty,+}^\times \det(K)\mathbb{A}_{\mathbb{Q}}^\times.$$

We let

$$X'_K := (\widetilde{X}_K/\mathcal{O}_{F,(p),+}^\times) \otimes \mathbb{F}.$$

We also have a natural action of $\mathcal{O}_{F,(p),+}^\times$ on the line bundle ω_β over $\widetilde{X}_K \otimes \mathbb{F}$ (induced by the identity on A). If K is sufficiently small (more precisely p-neat in the sense of [5, Def. 3.2.3]), then this action defines descent data to a line bundle on X'_K, which we also denote ω_β. Similarly we write ω^κ for the line bundle $\otimes_{\beta\in\mathbb{B}}\omega_\beta^{k_\beta}$ for $\kappa = \sum k_\beta e_\beta \in \mathbb{Z}^{\mathbb{B}}$. It is immediate from their definition that the partial Hasse invariants $\mathbf{h}_\beta$ (on the union of the $X_K^\epsilon \otimes \mathbb{F}$) descend to sections of ω^{h_β} over X'_K, and so also do the stratifications they define. For $T \subset \mathbb{B}$, we denote the resulting closed subscheme of X'_K by $X'_{K,T}$, or simply X'_T when K is clear from the context.

Similarly, the line bundles $\wedge^2\mathcal{H}_\beta$ on $\widetilde{X}_K \otimes \mathbb{F}$ descend to ones on X'_K, which we denote by δ_β. Note though that the canonical trivialization defined by ϵ_β does *not* descend to X'_K. However, we have canonical isomorphisms $\wedge^2\mathcal{H}_{\sigma^{-1}\beta} \xrightarrow{\sim} \wedge^2\mathcal{H}_\beta$, descending to isomorphisms $\delta_{\sigma^{-1}\beta}^p \xrightarrow{\sim} \delta_\beta$ for each $\beta \in \mathbb{B}$ (see for example [4, §3.2.2]). Writing δ^κ for the line bundle $\otimes_{\beta\in\mathbb{B}}\delta_\beta^{k_\beta}$, where $\kappa = \sum k_\beta e_\beta \in \mathbb{Z}^{\mathbb{B}}$, we thus obtain a canonical trivialization of δ^{h_β} for each β. It follows that δ^κ depends only on the character $(\mathcal{O}_F/p)^\times \to \mathbb{F}^\times$ induced by κ. More precisely, let Δ denote the group of characters $(\mathcal{O}_F/p)^\times \to \mathbb{F}^\times$, and consider the surjective homomorphism $\mathbb{Z}^{\mathbb{B}} \to \Delta$ defined by $\kappa \to \overline{\kappa}$ (induced by $e_\beta \mapsto \overline{\beta}$). One sees easily that the kernel is the lattice in $\mathbb{Z}^{\mathbb{B}}$ freely generated by the h_β, so that δ^κ depends (up to canonical isomorphism) only on $\overline{\kappa} \in \Delta$, and so we denote it $\delta^{\overline{\kappa}}$. Examining the definition of the section $\mathbf{b}_\beta(T)$ on (the union over ϵ of the) X_T, we see that it descends to a trivialization of $\delta^{-\overline{\beta}}\omega^{b_\beta}$ on X'_T, which we again denote $\mathbf{b}_\beta(T)$.

10.2. **Hecke actions and equivariance.** Recall from [4, §§2.1.3, 3.1.1] that if K_1 and K_2 are sufficiently small open subgroups of $\mathrm{GL}_2(\mathbb{A}_F^\infty)$ as above, and $g \in \mathrm{GL}_2(\mathbb{A}_F^{\infty,p})$ is such that $g^{-1}K_2^p g \subset K_1^p$, then we have a morphism $\rho_g : X'_{K_2} \to X'_{K_1}$ of schemes over $\mathbb{F}$, and isomorphisms $\rho_g^*\omega_\beta \xrightarrow{\sim} \omega_\beta$ and $\rho_g^*\delta_\beta \xrightarrow{\sim} \delta_\beta$ of line bundles over X_{K_2}, giving rise to an action of $\mathrm{GL}_2(\mathbb{A}_F^{\infty,p})$ on

$$\varinjlim_{K^p} H^0(X'_K, \delta^\xi\omega^\kappa)$$

for any $\xi \in \Delta$, $\kappa \in \mathbb{Z}^{\mathbb{B}}$. Identifying $H^0(X'_K, \delta^\xi\omega^\kappa)$ with the K^p invariants under this action, we thus also obtain an action of commuting Hecke operators T_v and S_v on $H^0(X'_K, \delta^\xi\omega^\kappa)$ for all primes v of F such that $\mathrm{GL}_2(\mathcal{O}_{F,v}) \subset K^p$. Furthermore, the partial Hasse invariants $\mathbf{h}_\beta$ are invariant under the action of $\mathrm{GL}_2(\mathbb{A}_F^{\infty,p})$, so the morphisms ρ_g restrict to morphisms $\rho_g : X'_{K_2,T} \to X'_{K_1,T}$ and give rise to an action of $\mathrm{GL}_2(\mathbb{A}_F^{\infty,p})$ on

$$\varinjlim_{K^p} H^0(X'_{K,T}, \delta^\xi\omega^\kappa).$$

We thus also obtain Hecke operators T_v and S_v on $H^0(X'_T, \delta^\xi\omega^\kappa)$ for all K and v as above, commuting with each other and with the injective maps

$$H^0(X'_T, \delta^\xi\omega^\kappa) \longrightarrow H^0(X'_T, \delta^\xi\omega^{\kappa+h_\beta})$$

defined by multiplication by $\mathbf{h}_\beta$ for $\beta \notin T$. Similarly, the isomorphisms

$$H^0(X'_T, \delta^\xi\omega^\kappa) \xrightarrow{\sim} H^0(X'_T, \delta^{\xi-\overline{\beta}}\omega^{\kappa+b_\beta})$$

defined by multiplication by $\mathbf{b}_\beta(T)$ for $\beta \in T$ are Hecke-equivariant.

We claim also that the sections $\mathbf{h}_\beta^{\beta'}(T)$ of Proposition 4.1 descend to Hecke-equivariant sections over X'_T. Indeed it follows exactly as in [4, §5.3] or [7, §6.5.2] that the isomorphisms constructed in the proof of [8, Thm. 5.2] are Hecke-equivariant in the following sense. Firstly, implicit in the construction of the isomorphism $\iota_{T'}$ is a choice of isomorphism $\alpha_v : \mathrm{GL}_2(F_v) \xrightarrow{\sim} B^\times_{S(T),v}$ for all finite $v \notin S(T)$ inducing an isomorphism $\mathrm{GL}_2(\mathbb{A}_F^{\infty,p}) \cong (B_{S(T)} \otimes_F \mathbb{A}_F^{\infty,p})^\times$. Furthermore, the isomorphisms α_v can be chosen (dependently on ϵ) so that the level K_T in the notation of §3 is $K'_T \cap G(\mathbb{A}^{\infty,p})$, where

$$K'_{T,p} = \prod_{\mathfrak{p}\in S(T)} \mathcal{O}^\times_{B,\mathfrak{p}} \prod_{\mathfrak{p}\in \mathrm{Iw}(T)} \alpha_\mathfrak{p}(K_0(\mathfrak{p})) \prod_{\mathfrak{p}\notin S(T)\cup \mathrm{Iw(T)}} \alpha_\mathfrak{p}(\mathrm{GL}_2(\mathcal{O}_{F,\mathfrak{p}})),$$

and the union over ϵ of the isomorphisms $\iota_{T'}$ descends to an isomorphism

$$X'_{T'} \xrightarrow[\cong]{\iota'_{T'}} \Pi_{\beta\in\tilde{T}-T'}\mathbb{P}(\mathcal{V}_\beta) \xrightarrow{\pi'_{T'}} X_{G',K'_T}$$

for a collection of rank two vector bundles $\mathcal{V}_\beta$ on the Shimura variety X_{G',K'_T} of level K'_T for the group $G' = B^\times_{S(T)}$. We thus obtain morphisms of strata $\pi'_{T,T'} : X'_T \to X'_{T'}$ exactly as in §3. Note also that for K_1, K_2 and g as above, we have $g_T^{-1}K'^p_{2,T}g_T \subset K'^p_{1,T}$ (writing g_T for $\alpha^p(g)$), and hence a morphism $\rho_{g_T} : X_{G',K'_{2,T}} \to X_{G',K'_{1,T}}$ of schemes over $\mathbb{F}$. Furthermore, the construction of the vector bundles $\mathcal{V}_\beta$ yields isomorphisms $\rho_{g_T}^*\mathcal{V}_\beta \xrightarrow{\sim} \mathcal{V}_\beta$, and hence a morphism

$$\prod_{\beta\in\widetilde{T}-T'} \mathbb{P}_{X_{G',K'_{2,T}}}(\mathcal{V}_\beta) \longrightarrow \prod_{\beta\in\widetilde{T}-T'} \mathbb{P}_{X_{G',K'_{1,T}}}(\mathcal{V}_\beta)$$

over ρ_{g_T}. The argument of [4, §5.3] shows that the isomorphisms $\iota_{T'}$ can be chosen (for all sufficiently small K) so that the resulting isomorphism $\iota'_{T'}$ is compatible with the above morphism of projectivizations for all $g \in \mathrm{GL}_2(\mathbb{A}_F^{\infty,p})$ (such that $g^{-1}K_2 g \subset K_1$). In particular, it follows that the morphisms $\pi'_{T,T'}$ between strata are compatible with the morphisms ρ_g. Furthermore, for $\beta \in \mathbb{B} - \widetilde{T}$, the line bundles η_β on (the union over ϵ of) the X_{G,K_T} descend to ones on X_{G',K'_T} such that $\iota'^*_{T'}\pi'^*_{T'}\eta_\beta \cong \omega_\beta$, so we obtain isomorphisms $\pi'^*_{T,T'}\omega^\kappa \cong \omega^\kappa$ exactly as in Lemma 3.2 and sections

$$\mathbf{h}_\beta^{\beta'}(T) \in H^0(X'_T, \omega^{h_\beta^{\beta'}})$$

exactly as in Proposition 4.1. Finally, for K_1, K_2 and g as above, we have canonical isomorphisms $\rho_{g_T}^*\eta_\beta \cong \eta_\beta$, which we see (as in [4, §5.4] or [7, §6.6]) are compatible under $\iota'^*_{T'}\pi'^*_{T'}\eta_\beta \cong \omega_\beta$ with the isomorphims $\rho_g^*\omega_\beta \cong \omega_\beta$. It follows that the sections $\mathbf{h}_\beta^{\beta'}(T)$ are invariant under the action of $\mathrm{GL}_2(\mathbb{A}_F^{\infty,p})$.

Recall that for $\beta \notin T$, we let

$$f_\beta(T) = \begin{cases} h^\beta_{\sigma^{n_\beta}\beta}, & \text{if } \beta \notin \widetilde{T}; \\ -b^\beta_{\sigma^{n_\beta}\beta}, & \text{if } \beta \in \widetilde{T} - T \end{cases}$$

(where slightly extend the definition of n_β to be $|\mathbb{B}_\mathfrak{p}|$ in the case $\mathbb{B}_\mathfrak{p} = \widetilde{T}_\mathfrak{p}$). We now define $\mathbf{f}_\beta(T)$ exactly as before, but as sections on X'_T, so that

- if $\beta \notin \widetilde{T}$, then $\mathbf{f}_\beta(T) = \mathbf{h}^\beta_{\sigma^{n_\beta}\beta} \in H^0(X'_T, \omega^{f_\beta(T)})$ if $\beta \notin \widetilde{T}$,
- and if $\beta \in \widetilde{T} - T$, then $\mathbf{f}_\beta(T) = \mathbf{h}^{\sigma\beta}_{\sigma^{n_\beta}\beta}\mathbf{b}_{\sigma\beta}(T)^{-p^{n_\beta-1}} \in H^0(X'_T, \delta^{\sigma^{n_\beta}\overline{\beta}}\omega^{f_\beta(T)})$ if $\beta \in \widetilde{T} - T$.

Note that by the discussion above, the $\mathbf{f}_\beta(T)$ are invariant under the action of $\mathrm{GL}_2(\mathbb{A}_F^{\infty,p})$. In particular, we have the following:

Proposition 10.1. *Suppose that $K = K_pK^p$ is a sufficiently small open subgroup of $\mathrm{GL}_2(\mathbb{A}_F^\infty)$ such that $K_p = \mathrm{GL}_2(\mathcal{O}_{F,p})$, and v is a prime of F such that $\mathrm{GL}_2(\mathcal{O}_{F,v}) \subset K^p \subset \mathrm{GL}_2(\mathbb{A}_F^{\infty,p})$. If $\beta \notin T \subset \mathbb{B}$, then*

1. *$T_v\mathbf{f}_\beta(T) = (\mathrm{Nm}_{F/\mathbb{Q}}v + 1)\mathbf{f}_\beta(T)$ and $S_v\mathbf{f}_\beta(T) = \mathbf{f}_\beta(T)$,*
2. *the map*

$$H^0(X'_T, \delta^\xi\omega^\kappa) \xrightarrow{f_\beta(T)\cdot} H^0(X'_T, \delta^{\xi'}\omega^{\kappa + f_\beta(T)})$$

commutes with the operators T_v and S_v, where $\xi' = \xi$ or $\xi + \sigma^{n_\beta}\overline{\beta}$ according to whether or not $\beta \notin \widetilde{T}$.

10.3. **Cones of weights.** Fix $T \subset \mathbb{B}$ and a sufficiently small open compact subgroup K of $\mathrm{GL}_2(\mathbb{A}_F^\infty)$ containing $\mathrm{GL}_2(\mathcal{O}_{F,p})$. Consider the cone of weights

$$C(X'_T, \delta^\bullet\omega^\bullet) = \{\, (\lambda, \kappa) \in \mathbb{Z}^\mathbb{B} \times \mathbb{Z}^\mathbb{B} \,:\, H^0(X'_T, \delta^\lambda\omega^\kappa) \neq 0 \,\}$$

and let $\mathcal{C}(X'_T, \delta^\bullet\omega^\bullet)$ be its $\mathbb{Q}^{\geq 0}$-saturation. We have the following immediate consequence of Theorem 7.6 and the discussion above.

Corollary 10.2. *We have $\mathcal{C}(X'_T, \delta^\bullet\omega^\bullet) = \mathbb{Q}^\mathbb{B} \times \mathcal{D}_T$.*

Proof. Suppose that $0 \neq \mathbf{f} \in H^0(X'_T, \delta^\lambda\omega^\kappa) \neq 0\,\}$. For each $\epsilon \in (\mathbb{A}_F^{\infty,p})^\times$, let

$$\mathbf{f}^\epsilon \in H^0(X^\epsilon_T, \delta^\lambda\omega^\kappa) \cong H^0(X^\epsilon_T, \omega^\kappa)$$

denote the pull-back of $\mathbf{f}$ via the finite étale morphism $X_T^\epsilon \to X'_T$ (where X_T^ϵ is the closed subscheme of $X_K^\epsilon \otimes \mathbb{F}$ defined by the vanishing of the $\mathbf{h}_\beta$ for $\beta \in T$). Since $\mathbf{f} \neq 0$, we have $\mathbf{f}^\epsilon \neq 0$ for some ϵ, and hence $\kappa \in \mathcal{C}_T = \mathcal{D}_T$ by Theorem 7.6. It follows that $\mathcal{C}(X'_T, \delta^\bullet\omega^\bullet) \subset \mathbb{Q}^{\mathbb{B}} \times \mathcal{D}_T$.

For the opposite inclusion, note that $C(X'_T, \delta^\bullet\omega^\bullet)$ contains each of the following:

- $\pm(h_\beta, 0)$ for all $\beta \in \mathbb{B}$ (from the canonical trivializations of δ^{h_β});
- $\pm(-e_\beta, b_\beta)$ for all $\beta \in T$ (from the trivializations $\mathbf{b}_\beta(T)$);
- $(0, f_\beta(T))$ for all $\beta \notin \widetilde{T}$ (from the sections $\mathbf{f}_\beta(T)$);
- $(e_{\sigma^{n_\beta}\beta}, f_\beta(T))$ for all $\beta \in \widetilde{T} - T$ (from the sections $\mathbf{f}_\beta(T)$).

Since the $\mathbb{Q}^{\mathbb{B}} = \langle h_\beta : \beta \in \mathbb{B}\rangle$ and $\mathcal{D}_T = \langle f_\beta(T) : \beta \notin T\rangle + \langle \pm b_\beta : \beta \in T\rangle$, it follows that $\mathbb{Q}^{\mathbb{B}} \times \mathcal{D}_T \subset \mathcal{C}(X'_T, \delta^\bullet\omega^\bullet)$. □

Corollary 10.3. *Suppose that $\xi \in \Delta$ and $\kappa \in \mathbb{Z}^{\mathbb{B}}$. If $\kappa \notin \mathcal{D}_T$, then $H^0(X'_T, \delta^\xi\omega^\kappa) = 0$.*

Proof. This is immediate from Corollary 10.2. □

We may also define a stratification on X'_T as in §8.1: For $J \subset \mathbb{B} - T$, we let

$$X'_{T,J} = X'_T \cap \big(\bigcap_{\beta\in J} Z(\mathbf{f}_\beta(T))\big), \quad \text{and} \quad W'_{T,J} = X'_{T,J} - \bigcup\nolimits_{\beta\notin J} X'_{T,\{\beta\}}.$$

Since the line bundles δ_β are torsion, the same proofs as before carry over to give the following:

Proposition 10.4. *For each $J \subset \mathbb{B} - T$, we have*

(1) *$W'_{T,J}$ is non-empty, quasi-affine and equidimensional of dimension $\mathbb{B} - (T \cup J)$;*
(2) *$X'_{T,J} = \overline{W}'_{T,J} = \bigcup_{J'\supset J} W'_{T,J'}$.*

Finally, we translate our results on minimal weights to the setting of $\mathrm{Res}_{F/\mathbb{Q}}\mathrm{GL}_2$. Suppose that $\kappa \in \mathbb{Z}^{\mathbb{B}}$, $\xi \in \Delta$, $0 \neq \mathbf{f} \in H^0(X'_T, \delta^\xi\omega^\kappa)$, and for each $\beta \notin T$, we let a_β be the maximal power of $\mathbf{f}_\beta(T)$ that divides $\mathbf{f}$ (which again exists by [3, Lemma 7.2]). By part (1) of Proposition 10.4, the strata $X'_T \cap Z(\mathbf{f}_\beta(T))$ and $X'_T \cap Z(\mathbf{f}_{\beta'}(T))$ have no common irreducible components for distinct $\beta, \beta' \notin T$, so $\mathbf{f}$ is divisible by $\prod_{\beta\in\mathbb{B}(T)} \mathbf{f}_\beta(T)^{a_\beta}$. We can therefore write

$$\mathbf{f} = \mathbf{f}_0 \prod_{\beta\notin T} \mathbf{f}_\beta(T)^{a_\beta}$$

for some $\mathbf{f}_0 \in H^0(X'_T, \delta^{\xi_0}\omega^{\kappa_0})$, where $\xi_0 = \xi - \sum_{\beta\in\widetilde{T}-T} a_\beta\sigma^{n_\beta}\overline{\beta}$, $\kappa_0 = \kappa - \sum_{\beta\notin T} a_\beta f_\beta(T)$, and $\mathbf{f}_0$ is not divisible by $\mathbf{f}_\beta(T)$ for any $\beta \notin T$.

Definition 10.5. For $0 \neq \mathbf{f} \in H^0(X'_T, \delta^\xi\omega^\kappa)$, we define the *minimal weight* of $\mathbf{f}$ to be

$$\Phi(\mathbf{f}) = i_T(\kappa_0) \in \mathbb{Z}^{\mathbb{B}-T},$$

where $i_T : \mathbb{Q}^{\mathbb{B}} \to \mathbb{Q}^{\mathbb{B}-T}$ is defined in (5.1), so in particular, $\Phi(\mathbf{f}) \in i_T(\mathcal{C}_T) = \mathcal{C}_T^{T-\mathrm{red}}$.

Proposition 10.6. *If $0 \neq \mathbf{f} \in H^0(X'_T, \delta^\xi\omega^\kappa)$, then $\Phi(\mathbf{f}) \in \mathcal{C}_T^{\min}$.*

Proof. Write $\mathbf{f} = \mathbf{f}_0\prod_{\beta\notin T}\mathbf{f}_\beta(T)^{a_\beta}$ as above, so $\mathbf{f}_0 \in H^0(X'_T, \delta^{\xi_0}\omega^{\kappa_0})$ and $i_T(\kappa_0) = \Phi(\mathbf{f})$. As in the proof of Proposition 9.2, if $\Phi(\mathbf{f}) \notin \mathcal{C}_T^{\min}$, then $L_{T,f_\beta,\tau}(\kappa_0) < 0$ for some $\beta \in \mathbb{B}(T)$ and $\tau \in \mathbb{B} - (T \cup \{\sigma^{n_\beta}\beta\})$.

For each $\epsilon \in (\mathbb{A}_F^{\infty,p})^\times$, let $\mathbf{f}_0^\epsilon \in H^0(X_T^\epsilon, \delta^{\xi_0}\omega^{\kappa_0})$ denote the pull-back of $\mathbf{f}_0$ to X_T^ϵ (as in the proof of Corollary 10.2). Since $L_{T,f_\beta,\tau}(\kappa_0) < 0$, it follows from Corollary 8.11 that $\mathbf{f}_0^\epsilon$ is divisible

by (the pull-back of) $\mathbf{f}_\beta(T)$. Since this holds for all ϵ, it follows that the pull-back of $\mathbf{f}_0$ to $\coprod_\epsilon X_T^\epsilon$ has the form $\mathbf{f}_1\mathbf{f}_\beta(T)$ for some

$$\mathbf{f}_1 \in H^0(\coprod_\epsilon X_T^\epsilon, \delta^{\xi_1}\omega^{\kappa_1}),$$

where $\xi_1 = \xi$ or $\xi_0 - \sigma^{n_\beta}\overline{\beta}$ and $\kappa_1 = \kappa_0 - f_\beta(T)$. Since $\mathbf{f}_0$ and $\mathbf{f}_\beta(T)$ descend to sections over X'_T, it follows that so does $\mathbf{f}_1$, contradicting the definition of $\mathbf{f}_0$. □

Combining Propositions 10.1 and 9.2 immediately gives the following:

Corollary 10.7. *Let S be a set of primes containing all $v|p$ and all v such that $\mathrm{GL}_2(\mathcal{O}_{F,v}) \not\subset K$. Suppose that $\mathbf{f} \in H^0(X'_T, \delta^\xi\omega^\kappa)$ is an eigenform for the Hecke operators T_v and S_v for all $v \notin S$. Then there is an eigenform $\mathbf{f}' \in H^0(X'_T, \delta^{\xi'}\omega^{\kappa'})$ with the same eigenvalues as $\mathbf{f}$ for T_v and S_v for all $v \notin S$, and such that $i_T(\kappa') \in \mathcal{C}_T^{\min}$.*

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